

Ratio & Proportion

Key - points:-

1) What are antecedent and consequent?

In the form $x : y$; x is called the *antecedent* and y is called *consequent*.

2) $x : y$ can also be written as $\frac{x}{y}$

3) Duplicate ratio (square) :-

Duplicate ratio of $x : y$ will be $x^2 : y^2$

4) Sub-duplicate Ratio (square root):-

Sub-duplicate Ratio of $x : y$ will be $\sqrt{x} : \sqrt{y}$

5) Triplicate Ratio (cube) :-

Triplicate ratio of $x : y$ will be $x^3 : y^3$

6) Sub-triplicate Ratio (cube root)

Sub-triplicate Ratio of $x : y$ will be $\sqrt[3]{x} : \sqrt[3]{y}$

7) Inverse ratio :- $x : y$ will be $y : x$

8) Compound Ratio : If two or more ratios are given, then the antecedent of one is multiplied with antecedent of the other and respective consequents are also multiplied.

If $a : b, c : d$ and $e : f$ are three ratios, then their compound ratio will be $\frac{ace}{bdf}$.

9) 3rd Proportional:- 3rd proportion of a and b i.e. $a : b :: b : x \Rightarrow$

$$\text{Then, } x = \frac{b^2}{a}$$

10) 4th Proportional: 4th proportional of a, b and c i.e. $a : b :: c : x$

$$\text{Then, } x = \frac{b \times c}{a}$$

11) Componendo and dividendo rule:

$$\text{If } a : b :: c : d, \text{ Then } \Rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}$$

12) Mean Proportion : Mean proportion of two numbers, a and b , is given by $\sqrt{ab}$.

Partnership

Concept 1

The ratio of profits of two or more partners will be equal to the ratio of their respective equivalent investment ratio.

Equivalent investment = (Amount Invested) $\times$ (Time for which the amount is invested).

We will take some examples to understand the concepts of partnership.

Example 1: A invests Rs. 10,000 and B invests Rs. 20,000 for a period of 1 year each. The profit, after 1 year will be divided in the ratio:

$$\frac{\text{Profit of A}}{\text{Profit of B}} = \frac{10000 \times 12}{20000 \times 12} = \frac{1}{2}$$

Example 2: A invests Rs.100 in a business. After 6 months B joins with an investment of Rs. 200. At the end of 1 year C joins with an investment of Rs.100. What will be the ratio of profit after 3 years?

	A	B	C
Amount(Rs.)	100	200	100
Time(month)	36	30	24
Equivalent Investment	3600	6000	2400
Profit Ratio	3	5	2

Example 3: In the previous case if the total profit after 3 years is 10,000. What will be the share of A?

Share of A in total profit =

$$\frac{3}{3+5+2} \times 10,000 = \frac{3}{10} \times 10000 \\ \Rightarrow 3 \times 1000 = 3000$$

Note: In case someone invests in - between the total period of time, just take care for what time an amount is invested. Similarly there can be a case when someone opts out in that case also, the time for which the amount was invested will be taken into consideration and not the full period.

Concept 2

If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \frac{f}{g} = K$, where K is constant. Then, $K = \frac{a+c+e+f}{b+d+f+g}$

Example : $\frac{1+x}{2-x} = \frac{9-x}{3+x} = K$, Find the value of K ?

$$\text{Sol. :- } K = \frac{1+x+9-x}{2-x+3+x} = \frac{10}{5} = 2$$

Concept 3

If two or three ratios are given, and we have to find an individual ratio of all of them, then we take LCM of the common entities and then balancing the common entities, we get the required ratio.

Example :- $a : b = 3 : 4, b : c = 6 : 7$.
LCM of the common entities i.e $b = (4, 6) = 12$

$$a : b = 3 : 4 \Rightarrow a : 12 = 9 : 12$$

$$b : c = 6 : 7 \Rightarrow 12 : 14 = 12 : 14$$

$$\text{So, } a : b : c = 9 : 12 : 14$$

Note : In case there are more than three ratios given, then the LCM approach will be tedious in finding the individual ratio of the given entities.

Alternative method

$$\text{If } a : b = 2 : 3, \quad b : c = 4 : 5$$

$$c : d = 6 : 7, \quad d : e = 1 : 2$$

$$\begin{array}{ccccccc} a & : & b & : & c & : & d & : & e \\ \hline 2 & : & 3 & & 3 & & 3 & & 3 \\ 4 & & 4 & : & 5 & & 5 & & 5 \\ 6 & & 6 & & 6 & : & 7 & & 7 \\ \hline 1 & & 1 & & 1 & & 1 & : & 2 \end{array}$$

$$48 : 72 : 90 : 105 : 210$$

To understand this concept we will take some examples

Example 1 : ₹5600 is to be divided among A, B, C & D in such a way that the ratio of share of A : B is 1 : 2, B : C is 3 : 1, C : D is 2 : 3, find the share of A and B taken together?

Share of A : B = 1 : 2

Share of B : C = 3 : 1

Share of C : D = 2 : 3

A : B : C : D $\rightarrow$

$$1 \times 3 \times 2 : 2 \times 3 \times 2 : 2 \times 1 \times 2 : 2 \times 1 \times 3 \\ = 6 : 12 : 4 : 6 \text{ or } 3 : 6 : 2 : 3$$

Sum of the share of all of them = $3 + 6 + 2 + 3 = 14$ unit

$$\text{Total share of (A + B)} = \frac{6+3}{14} \times 5600 \\ = 9 \times 400 = ₹3600$$

Example 2 : Sum of ₹ x is divided among A, B, C such that the ratio of the shares of A and B is 6 : 7 and that of B and C is 3 : 2. If the difference of the share of A and C is ₹540. Find the value of x

Share of A and B = 6 : 7

Share of B and C = 3 : 2

Ratio of the shares of A, B and C = $6 \times 3 : 3 \times 7 : 7 \times 2 = 18 : 21 : 14$

Diff. of A and C = $18 - 14 = 4$ unit, which corresponds to ₹540

$$\text{Total share of A, B and C} = 18 + 21 + 14 \\ = 53 \text{ unit} = \frac{540}{4} \times 53 = ₹7155$$

Concept 4

If numerator and denominator of a ratio is multiplied and divided by same number, the ratio remains unchanged.

$$\text{E.g. :- } \frac{a}{b} = \frac{a \times k}{b \times k} = \frac{\frac{a}{k}}{\frac{b}{k}}$$

Ratio is independent of change in scale only and not of origin.

Concept 5

If there is 'n' number of unequal fractions such as $\frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3}, \frac{a_4}{b_4}, \dots, \frac{a_n}{b_n}$ then the ratio of $(a_1 + a_2 + a_3 + \dots + a_n)$ and $(b_1 + b_2 + b_3 + \dots + b_n)$ lies between lowest and highest values of these fractions.

Example:- three unequal fractions are $\frac{1}{2}, \frac{2}{3}, \frac{4}{7}$. So, the ratio of $(1 + 2 + 4) : (2 + 3 + 7) = 7 : 12$. Among these fractions,

lowest fraction is $\frac{1}{2}$ and the highest fraction is $\frac{2}{3}$ so, the required value lies between $\frac{1}{2}$ and $\frac{2}{3}$

Concept 6:

If $aA = bB = cC$, where a, b, c are constants and A, B, C are variables. Then the ratio of $A : B : C = b \times c : c \times a : a \times b$.

Example: if $3A = 4B = 5C$, then find $A:B:C$
The ratio of $A : B : C = 4 \times 5 : 5 \times 3 : 3 \times 4 = 20 : 15 : 12$

Variety Questions

Q.1. A bag contains ₹5, ₹2, and ₹1 coins in the ratio 2 : 3 : 5. If the total amount of money in the bag is ₹252, find the number of coins of each kind.
SSC CGL 18/09/2025 (Shift 1)
(a) 16 coins of ₹5, 24 coins of ₹2, and 40 coins of ₹1
(b) 20 coins of ₹5, 30 coins of ₹2, and 50 coins of ₹1
(c) 24 coins of ₹5, 36 coins of ₹2, and 60 coins of ₹1
(d) 18 coins of ₹5, 27 coins of ₹2, and 45 coins of ₹1

Q.2. If $9A = 10B$ and $12B = 13C$, find the ratio $A : C$
Higher Secondary 29/07/2025 (Shift - 2)
(a) 15 : 16 (b) 39 : 40
(c) 27 : 28 (d) 65 : 54

Q.3. The savings of two persons are in the ratio 8 : 7. If the first person increases his savings by 20% and the second person increases his savings by 15%, find the new ratio of their savings.
Higher Secondary 28/07/2025 (Shift - 2)
(a) 32 : 28 (b) 46 : 40.25
(c) 9.6 : 8.05 (d) 184 : 161

Q.4. A scholarship amount of ₹1500 is to be distributed between A and B in the ratio 7:5. An amount of ₹250 is deducted for administrative charges; the remaining amount is increased by 20% as a bonus before the final distribution. How much amount will student A receive?
Higher Secondary 24/07/2025 (Shift - 2)
(a) ₹875 (b) ₹900 (c) ₹833.33 (d) ₹800

Q.5. In a zoo, there are horse and pigeons. If heads are counted, there are 200 heads and if legs are counted there are 580 legs. How many pigeons are there?
Matriculation Level 29/07/2025(Shift - 1)
(a) 90 (b) 110 (c) 100 (d) 120

Q.6. A recipe needs 4 cups of sugar to

make 16 cookies. How many cups of sugar are needed to make 24 cookies in the same proportions?

Matriculation Level 24/07/2025(Shift - 1)

- (a) 5 cups (b) 8 cups
(c) 6 cups (d) 7 cups

Q.7. The number of marbles in three bags is in the ratio of 2 : 3 : 4. If 12 marbles are added to each bag, then the ratio changed to 8 : 11 : 14. The total number of marbles in the beginning was:
(a) 216 (b) 162 (c) 200 (d) 126

Q.8. A function of two variable varies directly with x and inversely with y . Determine function when $x = 5$ and $y = 3$. Given that, for $x = 0$ and $y = 1$, $f = 15$ and for $x = 1$ and $y = 15$, $f = 2$.
(a) 10 (b) 2 (c) 5 (d) 15

Q.9. A payment of ₹120 is made with ₹10, ₹5 and ₹2 coins. A total of 25 coins are used. Which of the following is the number of ₹10 coins used in the payment?
(a) 6 (b) 4 (c) 10 (d) 8

Q.10. What is the ratio of the mean proportional of 1.21 and 0.09 with the mean proportional of 0.16 and 0.36 ?
(a) 11 : 8 (b) 13 : 8 (c) 13 : 12 (d) 11 : 13

Q.11. The mean proportional of $8a^2$ and $\frac{18}{a^4}$, where a is a positive number, is ____.
(a) $\frac{12}{a^2}$ (b) $12a$ (c) $12a^2$ (d) $\frac{12}{a}$

Q.12. R pays ₹100 to P with ₹5, ₹2 and ₹1 coins. The total number of coins used for paying are 40. What is the number of coins of denomination ₹5 in the payment?
(a) 16 (b) 17 (c) 18 (d) 13

Q.13. Find the fourth proportion of the following numbers: $\frac{1}{3}$ rd of 27, $\frac{4}{5}$ th of 30, $\frac{3}{7}$ th of 35.
(a) 35 (b) 40 (c) 45 (d) 38

Q.14. The concentration of three wines A, B and C are 10%, 20% and 30%, respectively. They are mixed in the ratio 2 : 3 : x resulting in a 23% concentration solution. Find the value of x .
(a) 7 (b) 6 (c) 8 (d) 5

Q.15. In a school of 720 students, the ratio of boys to girls is 3 : 5. Find how many new boys may be allowed to the school if 18 new girls are admitted, so that the ratio of boys to girls changes to 2 : 3.

- (a) 42 (b) 50 (c) 44 (d) 38

Q.16. Ram has 12 hens and 5 cows, Shyam has 18 hens and 9 cows, Preetham has 4 hens and 8 cows, and Vinay has 10 hens and 7 cows. Who has a greater ratio of hens to cows?
(a) Shyam (b) Preetham
(c) Vinay (d) Ram

Q.17. A company is manufacturing boxes of three types - plastic, wood and glass. The ratio of the numbers of plastic, wood and glass boxes is 2 : 3 : 5. The difference between the numbers of wood and glass boxes is 144. If the number of plastic boxes is increased by 25%, then what will be the number of plastic boxes now ?
(a) 144 (b) 216 (c) 360 (d) 180

Q.18. Find the fourth proportional of 12^{35} , 3^{12} and 4^{12} .
(a) 12^{-23} (b) 3^{24} (c) 12^{-11} (d) 4^{24}

Q.19. Three numbers are in the ratio 1 : 2 : 3. By adding 5 to each of the numbers then the ratio changes to 2 : 3 : 4. The smallest number is:
(a) 5 (b) 15 (c) 12 (d) 20

Q.20. When $x = 1$ then y is directly proportional to $x + 3$ and $y = 8$. When $z = 6$ then x is inversely proportional to z and $x = 2$. Find the relation between y and z .
(a) $(y - 6) / z = 24$ (b) $(y - 6) / z = 12$
(c) $z(y - 6) = 12$ (d) $z(y - 6) = 24$

Q.21. If $A : B = \frac{1}{3} : \frac{1}{5}$, $B : C = \frac{1}{7} : \frac{1}{8}$ and $C : D = \frac{1}{10} : \frac{1}{21}$, then $A : B : C : D$ is:
(a) 40 : 24 : 21 : 10 (b) 48 : 23 : 25 : 8
(c) 8 : 3 : 7 : 5 (d) 14 : 24 : 28 : 10

Q.22. The average of the two numbers is 48. If 3 is added to a smaller number, then the ratio between them is 11 : 20. Find the larger number. (Round up to two decimal places).
(a) 40.87 (b) 3.87 (c) 63.87 (d) 70.87

Q.23. Arrange the ratios in ascending order $5 : 3\frac{1}{3}$, $6\frac{1}{4} : 3\frac{1}{8}$ and $3.5 : 4.5$.
(a) $3.5 : 4.5 < 5 : 3\frac{1}{3} < 6\frac{1}{4} : 3\frac{1}{8}$
(b) $6\frac{1}{4} : 3\frac{1}{8} < 5 : 3\frac{1}{3} < 3.5 : 4.5$
(c) $5 : 3\frac{1}{3} < 3.5 : 4.5 < 6\frac{1}{4} : 3\frac{1}{8}$
(d) $5 : 3\frac{1}{3} < 6\frac{1}{4} : 3\frac{1}{8} < 3.5 : 4.5$

Q.24. The incomes of P, Q and R are in the ratio of 9 : 15 : 10 and their savings



are in the ratio is 10 : 15 : 12. If R spends 60% of his income, what is the ratio of the expenditures of P, Q and R?

- (a) 18 : 17 : 30 (b) 17 : 30 : 18
(c) 17 : 18 : 30 (d) 18 : 30 : 17

Q.25. Find the fourth proportional to $(a + b)^2 \times (a^3 - b^3), (a^3 + b^3), (a^2 - b^2)$.

- (a) $\frac{(a^2 - ab + b^2)}{(a + b)}$ (b) $\frac{(a^2 + ab + b^2)}{(a - b)}$
(c) $\frac{(a^2 - ab + b^2)}{(a^2 + ab + b^2)}$ (d) $\frac{(a^2 + ab + b^2)}{(a^2 - b^2)}$

Q.26. Two numbers A and B are such that the sum of 6% of A and 4% of B is $\frac{2}{5}$

of the sum of 7% of A and 11% of B, then the ratio A : B is:

- (a) 3 : 7 (b) 5 : 8 (c) 1 : 8 (d) 1 : 7

Q.27. A certain sum was to be divided between A and B in the ratio 8 : 5. However, by mistake, it was divided in the ratio 5 : 8. Thus, A gets ₹300 less than his original share. Find the sum.

- (a) ₹2,600 (b) ₹1,500 (c) ₹1,300 (d) ₹600

Q.28. When P is subtracted from each of the numbers 8, 6, 2 and 9, the numbers so obtained in this order are in proportion. What is the mean proportional between $(3P - 6)$ and $(9P - 4)$?

- (a) 26 (b) 28 (c) 29 (d) 24

Q.29. A sum of money is divided among A, B, C in the ratio 3 : 2 : 5, respectively. If the share of C is ₹300 more than B. Then how much more money will A get than B?

- (a) ₹150 (b) ₹75 (c) ₹100 (d) ₹125

Q.30. The mean proportional of

$(a + b)(a - b)^3, (a + b)^3(a - b)$ is:

- (a) $(a + b)^2(a - b)^2$ (b) $(a + b)^2(a - b)$
(c) $(a + b)(a - b)^2$ (d) $a^2 - b^2$

Q.31. The seats for Mathematics, English and Chemistry in a school are in the ratio of 6 : 4 : 7. If these seats are increased by 20%, 40% and 60%, respectively, the ratio of the increased seats is:

- (a) 9 : 7 : 8 (b) 9 : 7 : 14
(c) 1 : 2 : 3 (d) 7 : 9 : 14

Q.32. A bag contains ₹550 in the form of 50p, 25p and 20p coins in the ratio 2 : 3 : 5. The difference between the amounts that are contributed by the 50 p and the 20 p coins is:

- (a) ₹30 (b) ₹20 (c) ₹10 (d) ₹0

Q.33. In a party hall, there are people in blue and red dresses. The ratio of the number of men in blue to the number of

women in red is 3 : 7. The ratio of men in red to the number of women in blue is 2 : 1. If the ratio of the number of people in blue to the number of people in red is 35 : 76, then what is the ratio of the number of men to the number of women?

- (a) 47 : 51 (b) 40 : 43
(c) 62 : 57 (d) 52 : 59

Q.34. P's income is 250 percent of Q's income and P's expenditure is 180 percent of Q's expenditure. If P's income is 400 percent of Q's expenditure, then what is the respective ratio of P's savings and Q's savings?

- (a) 8 : 3 (b) 11 : 3 (c) 11 : 5 (d) 5 : 2

Q.35. The number of units manufactured by company A was 12500 units in 2019 and 10625 units in 2020. While in company B, the production fell from 34000 units in 2019 to 30600 units in 2020. If X and Y are the percentage decrease in the number of units manufactured by company A and B respectively from 2019 to 2020, then what will be the ratio of X and Y?

- (a) 5 : 3 (b) 3 : 4 (c) 3 : 2 (d) 8 : 5

Q.36. In an office with 40 officers, the average salaries of class-A, class-B and class-C officers are ₹600, ₹750 and ₹1,000 a day, respectively. The number of class-A, class-B and class-C officers in the office are in the ratio 5 : 4 : 1, respectively. Find the monthly average salary (in ₹) of an officers, all the 40 of them taken together. [Assume the number of days in a month to be 30.]

- (a) 25,800 (b) 21,000
(c) 24,030 (d) 24,600

Q.37. The ratio of income of A to that of B is 4 : 3. The expenditure of A is $\frac{5}{12}$ of his income and the expenditure of B is $\frac{7}{15}$ of his income. The difference between their expenditure is ₹4,500. Find the income of B.

- (a) ₹49,625 (b) ₹48,625
(c) ₹51,625 (d) ₹50,625

Q.38. A sum of ₹46,800 is divided among A, B, C and D in such a way that the ratio of the combined share of A and D to the combined share of B and C is 8 : 5. The ratio of the share of B to that of C is 5 : 4. A receives ₹18,400. If x is the difference between the shares of A and B and y is the difference between the shares of C and D, then what is the value of $(x - y)$ (in ₹)?

- (a) 6000 (b) 6500 (c) 5000 (d) 7000

Q.39. If a number of pens and pencils are bought in the ratio of 5 : 3, then Ashok has to pay ₹44. If the ratio is changed as 3 : 5, then he has to pay ₹36. Find the ratio of the price of one pen to one pencil.

- (a) 5 : 3 (b) 7 : 3 (c) 8 : 5 (d) 7 : 4

Q.40. Atul purchased Bread costing Rs.20 and gave a 100 rupee note to the shopkeeper. The shopkeeper gave the balance money in coins of denomination 2, 5, and 10. If these coins are in the ratio 5 : 4 : 1, then how many 5 coins did the shopkeeper give?

- (a) 5 (b) 6 (c) 8 (d) 4

Q.41. In a school, $\frac{3}{8}$ of the number of students are girls and the rest are boys, One - third of the number of boys are below 10 years and $\frac{2}{3}$ of the number of girls is also below 10 years. If the number of students of age 10 or more years is 260, then the number of boys in the school is:

- (a) 280 (b) 300 (c) 234 (d) 312

Q.42. By adding 3 and 5 in numerator and denominator of a fraction it becomes $\frac{2}{3}$. If 1 and 3 are subtracted and added from numerator and denominator respectively, it becomes $\frac{2}{5}$. Find the fraction.

- (a) $\frac{5}{7}$ (b) $\frac{6}{7}$ (c) $\frac{7}{6}$ (d) $\frac{7}{5}$

Q.43. A, B and C spend 80%, 85% and 75% of their incomes, respectively. If their savings are in the ratio 8 : 9 : 20 and the difference between the incomes of A and C is Rs. 18,000, then the income of B is:

- (a) Rs. 24,000 (b) Rs. 27,000
(c) Rs. 30,000 (d) Rs. 36,000

Q.44. The ratio of income of A and B last year was 4 : 3, respectively. The ratios of their individual incomes of the last year and the present year are 3 : 4 and 5 : 6, respectively. If their total income for the present year is Rs. 8.04 lakh, then the income of B last year was:

- (a) Rs. 2.7 lakh (b) Rs. 3.6 lakh
(c) Rs. 2.4 lakh (d) Rs. 2.8 lakh

Q.45. The ratio of the income of A to that of B is 5 : 7. A and B save Rs. 4000 and Rs. 5000 respectively. If the expenditure of A is equal to $66\frac{2}{3}\%$ of the expenditure of B, then the total income of A and B is:



- (a) Rs. 25,200 (b) Rs. 24,000
(c) Rs. 26,400 (d) Rs. 28,800

Q.46. The ratio between the divisor and remainder is 3 : 2 and the ratio between the divisor and the quotient is 7 : 12. If the remainder is 14, then find the remainder when the dividend is divided by 9.

- (a) 3 (b) 6 (c) 4 (d) 5

Q.47. Ravi's age is $\frac{3}{5}$ of Shyam's age.

After x years the ratio of the ages of Ravi and Shyam becomes 5 : 7. If initially the sum of their ages is 32, then what is the value of x ?

- (a) 2 (b) 4 (c) 8 (d) 7

Q.48. Rs. 8000 is distributed among A, B and C such that they receive notes of Rs. 500, Rs. 200 and Rs. 100 respectively. The amounts received by them are in the ratio 15 : 2 : 3. What was the ratio of the numbers of notes of Rs. 500, Rs 200 and Rs 100 ?

- (a) 3:1:3 (b) 3:3:1 (c) 4:1:2 (d) 3:2:2

Q.49. A sum of Rs. 4360 was to be divided among A, B, C and D in the ratio of 3 : 4 : 5 : 8, but it was divided in the ratio of $\frac{1}{3} : \frac{1}{4} : \frac{1}{5} : \frac{1}{8}$ by mistake.

As a result:

- (a) A received Rs. 956 more
(b) B received Rs. 318 more
(c) D received Rs. 1144 less
(d) C received Rs. 132 less

Q.50. A sum of Rs. x is divided among A, B and C such that the ratio of shares of A and B is 7 : 12 and that of B and C is 8 : 5. If the difference in the shares of A and C is Rs. 214, then the value of x is :

- (a) 11556 (b) 11128 (c) 11770 (d) 11342

Q.51. The prices of two articles are in the ratio 4 : 5. If the price of the first article is increased by x% and that of the other is decreased by 30%, then the new prices of A and B will be in the ratio 10 : 7. The value of x is :

- (a) 24.5 (b) 22.5 (c) 25 (d) 20

Examination wise Questions

SSC CGL 2025 Tier - I

Q.52. The number of red, blue, and green marbles in a bag is in the ratio 3 : 4 : 6. If 25 red marbles, 15 blue marbles, and an unknown number of green marbles are added to the bag, the ratio of red, blue, and green marbles becomes 4 : 5 : 9.

Determine the number of green marbles added.

SSC CGL 12/09/2025 (Shift 2)

- (a) 103 (b) 110 (c) 107 (d) 105

Q.53. The ratio of boys to girls in a school is 5 : 3. If 10 more boys join the school, the new ratio of boys to girls becomes 2:1. What is the total number of students in the school initially?

SSC CGL 12/09/2025 (Shift 2)

- (a) 50 (b) 80 (c) 60 (d) 100

Q.54. If 50% of A = 0.4 of B = $\frac{1}{8}$ of C,

find A : B : C

SSC CGL 12/09/2025 (Shift 3)

- (a) 4 : 5 : 16 (b) 16 : 10 : 4
(c) 6 : 3 : 10 (d) 10 : 4 : 16

Q.55. If B is 30% more than A, and C is 25% more than B, then what is A : C ?

SSC CGL 13/09/2025 (Shift 3)

- (a) 13 : 8 (b) 8 : 13 (c) 16 : 25 (d) 25 : 16

Q.56. A rope is divided in the ratio $\frac{2}{5}$:

$\frac{1}{4} : \frac{3}{5} : 1$. Total length = 5 m. Find the smallest piece.

SSC CGL 13/09/2025 (Shift 3)

- (a) 0.556 m (b) 0.889m
(c) 1.333m (d) 2.222m

Q.57. If M : N = 3 : 4, N : O = 5 : 6, and O : P = 8 : 10, then find N : P.

SSC CGL 14/09/2025 (Shift 2)

- (a) 5 : 7 (b) 2 : 3 (c) 4 : 5 (d) 3 : 2

Q.58. What should be subtracted from 52, 47, 20, 19 so that the remaining numbers may be proportional ?

SSC CGL 14/09/2025 (Shift 2)

- (a) 3 (b) 9 (c) 4 (d) 12

Q.59. The number of red, blue, and green balls in a bag is in the ratio 2: 3: 5. If 10 red balls, 10 blue balls, and an unknown number of green balls are added to the bag, the ratio of red, blue, and green balls becomes 3 : 4 : 6. Determine the number of green balls added.

SSC CGL 14/09/2025 (Shift 3)

- (a) 3 (b) 6 (c) 7 (d) 10

Q.60. If P : Q = 7 : 5, Q : R = 2 : 3, and R : S = 4 : 5, find the ratio P : S.

SSC CGL 15/09/2025 (Shift 1)

- (a) 28 : 25 (b) 56 : 75
(c) 30 : 25 (d) 70 : 56

Q.61. If x : y = 2 : 5, then find $(3x + 2y) : (7x - 2y)$.

SSC CGL 15/09/2025 (Shift 2)

- (a) 3 : 2 (b) 4 : 1 (c) 7 : 4 (d) 9 : 5

Q.62. A ₹48,000 annual bonus is

distributed among Sales, Marketing, and Operations in a 5 : 4 : 3 ratio. Marketing uses ₹3,000 for team building and ₹1,500 for an intern, then allocates the rest to two top performers in a 3 : 2 ratio. How much does the top performer with the smaller share receive ?

SSC CGL 15/09/2025 (Shift 3)

- (a) ₹5,833.33 (b) ₹5,250
(c) ₹4,600 (d) ₹4,200

Q.63. If A : B = 4 : 7, B : C = 3 : 5, and C : D = 8 : 9, find A : B : C : D.

SSC CGL 16/09/2025 (Shift 1)

- (a) 96 : 168 : 280 : 315
(b) 168 : 96 : 280 : 315
(c) 96 : 168 : 315 : 280
(d) 96 : 280 : 168 : 315

Q.64. A box contains ₹10, ₹5, and ₹2 coins in the ratio 1 : 2 : 3. If the total amount of money in the box is ₹416, find the number of coins of each kind.

SSC CGL 16/09/2025 (Shift 1)

- (a) 20 coins of ₹10, 40 of ₹5, and 60 of ₹2
(b) 16 coins of ₹10, 32 of ₹5, and 48 of ₹2
(c) 10 coins of ₹10, 20 of ₹5, and 30 of ₹2
(d) 16 coins of ₹10, 32 of ₹5, and 46 of ₹2

Q.65. If a : b = 5 : 3, find $(3a + 4b) : (3a - 4b)$.

SSC CGL 16/09/2025 (Shift 2)

- (a) 9 : 1 (b) 1 : 9 (c) 7 : 1 (d) 1 : 7

Q.66. If 20% of a is equal to 15% of b, then find the ratio a : b.

SSC CGL 16/09/2025 (Shift 3)

- (a) 6 : 5 (b) 3 : 4 (c) 2 : 5 (d) 5 : 4

Q.67. What should be subtracted from 5, 8, 7, 12 so that the remaining numbers may be proportional ?

SSC CGL 16/09/2025 (Shift 3)

- (a) 2 (b) 3 (c) 4 (d) 5

Q.68. Given that W : X = 3 : 5, X : Y = 5 : 7, and Y : Z = 7 : 9, find the compound ratio W : X : Y : Z.

SSC CGL 17/09/2025 (Shift 1)

- (a) 20 : 24 : 56 : 63 (b) 3 : 5 : 7 : 9
(c) 5 : 6 : 14 : 9 (d) 20 : 24 : 42 : 63

Q.69. If 40% of $(P + Q) = 60\%$ of $(P - Q)$, then find the ratio P : Q.

SSC CGL 17/09/2025 (Shift 2)

- (a) 5 : 1 (b) 2 : 3 (c) 2 : 5 (d) 3 : 2

Q.70. If P is 25% more than Q, and Q is 20% more than R, then what is P : R ?

SSC CGL 17/09/2025 (Shift 3)

- (a) 3 : 2 (b) 4 : 3 (c) 7 : 4 (d) 3 : 4

Q.71. If 20% of $(A + B) = 60\%$ of $(A - B)$,



then find the ratio A : B.

SSC CGL 18/09/2025 (Shift 1)

(a) 1 : 2 (b) 2 : 1 (c) 1 : 3 (d) 3 : 1

Q.72. If 44% of P is equal to 24% of Q, what is the ratio P : Q ?

SSC CGL 18/09/2025 (Shift 2)

(a) 6 : 11 (b) 6 : 13 (c) 13 : 7 (d) 11 : 6

Q.73. If 20% of A = 0.5 of B = $\frac{1}{4}$ of C,

then what is A : B : C ?

SSC CGL 18/09/2025 (Shift 3)

(a) 5 : 2 : 4 (b) 5 : 4 : 2
(c) 5 : 4 : 8 (d) 5 : 3 : 2

Q.74. If 75% of P = 60% of Q = 30% of R, find P : Q : R.

SSC CGL 19/09/2025 (Shift 1)

(a) 4 : 5 : 10 (b) 10 : 5 : 4
(c) 5 : 4 : 10 (d) 2 : 3 : 5

Q.75. If 37.5% of p is equal to 25% of q, what is the ratio p : q ?

SSC CGL 19/09/2025 (Shift 2)

(a) 1 : 4 (b) 4 : 1 (c) 3 : 2 (d) 2 : 3

Q.76. If $(A + B) : (A - B) = 7 : 3$, find $(A^3 + B^3) : (A^3 - B^3)$

SSC CGL 19/09/2025 (Shift 3)

(a) 195 : 148 (b) 125 : 137
(c) 133 : 117 (d) 123 : 117

Q.77. A wire is cut into three pieces in the ratio 3 : 4 : 8. If the length of the longest piece is 40 cm, what is the total length of the wire ?

SSC CGL 20/09/2025 (Shift 1)

(a) 40 cm (b) 50 cm (c) 75 cm (d) 85 cm

Q.78. Three friends share a winning prize in the ratio 3 : 4 : 5. If the total prize is Rs. 15000, how much more does the person with the highest share get than the lowest ?

SSC CGL 21/09/2025 (Shift 2)

(a) 3500 (b) 3000 (c) 2500 (d) 1250

Q.79. Two numbers are in the ratio 3 : 5. If the difference between them is 16, find the numbers.

SSC CGL 22/09/2025 (Shift 1)

(a) 24 and 40 (b) 18 and 35
(c) 15 and 35 (d) 18 and 25

Q.80. The weights of three friends are in the ratio 4 : 5 : 6. If each gains 9 kg, the new ratio becomes 13 : 14 : 15. What is the original weight of the heaviest friend?

SSC CGL 22/09/2025 (Shift 2)

(a) 5 kg (b) 7 kg (c) 6 kg (d) 9 kg

Q.81. If $x : y = 3 : 7$, then what is $(2x + 3y) : (x + y)$?

SSC CGL 22/09/2025 (Shift 3)

(a) 27 : 10 (b) 7 : 19 (c) 16 : 7 (d) 7 : 16

Q.82. A 30-liter solution contains acid and water in the ratio 2 : 1. How much water must be added to make the ratio 1 : 1?

SSC CGL 23/09/2025 (Shift 2)

(a) 11 L (b) 10 L (c) 12 L (d) 21 L

Q.83. P and Q share profit in 9 : 7. If P gets Rs. 27,000, how much did Q receive?

SSC CGL 24/09/2025 (Shift 1)

(a) Rs. 21,500 (b) Rs. 10,000
(c) Rs. 21,000 (d) Rs. 22,000

Q.84. $P : Q = 2 : 3$, $Q : R = 6 : 5$, $R : S = 10 : 7$. What is the ratio P : S ?

SSC CGL 24/09/2025 (Shift 1)

(a) 8 : 7 (b) 5 : 6 (c) 11 : 13 (d) 7 : 8

Q.85. A box has 120 pencils in red, blue, and black in the ratio 3:2:5. If 10 red and 20 black pencils are removed, what is the new ratio of red to black pencils?

SSC CGL 24/09/2025 (Shift 3)

(a) 1 : 2 (b) 3 : 5 (c) 13 : 15 (d) 13 : 20

Q.86. If 50% of A = $\frac{1}{4}$ of B = $\frac{1}{6}$ of C,

find A : B : C

SSC CGL 14/10/2025 (Shift 1)

(a) 1 : 2 : 3 (b) 11 : 12 : 13
(c) 6 : 4 : 5 (d) 7 : 4 : 8

Q.87. The weights of three friends are in the ratio 8 : 7 : 3. If each gains 9 kg, the new ratio becomes 17 : 16 : 12. What is the original weight of the heaviest friend?

SSC CGL 14/10/2025 (Shift 1)

(a) 4 kg (b) 8 kg (c) 10 kg (d) 7 kg

Q.88. If a number x is subtracted from 52, 47, 20, and 19, the resulting numbers are in proportion. What is the value of x + 5 ?

SSC CGL 14/10/2025 (Shift 3)

(a) 13 (b) 19 (c) 17 (d) 12

Q.89. A stationery shop ordered 16 dozen black pens and some additional dozen blue pens. The price of the black pens per dozen was two times that of the blue ones. When the order was delivered, it was found that the number of dozens of the two colours had been interchanged. This increased the bill by 50%. Find the ratio of the number of dozens of black pens to the number of dozens of blue pens in the original order.

SSC CGL 14/10/2025 (Shift 3)

(a) 1 : 4 (b) 1 : 2 (c) 2 : 3 (d) 3 : 4

Q.90. If 75% of P = 60% of Q = 30% of R, find P : Q : R.

SSC CGL 14/10/2025 (Shift 3)

(a) 4 : 5 : 10 (b) 10 : 5 : 4
(c) 5 : 4 : 10 (d) 2 : 3 : 5

SSC Selection Post (XIII)

Q.91. In a village, there are 3,200 females for every 4,000 males. What is the sex ratio in this city (number of females per 1000 males) ?

Matriculation Level 24/07/2025(Shift - 1)

(a) 700 (b) 750 (c) 820 (d) 800

Q.92. A jar contains marbles in the ratio 4 : 6 : 5 (red : green : blue). If there are 150 marbles in total, how many are green?

Matriculation Level 25/07/2025(Shift - 1)

(a) 40 (b) 45 (c) 60 (d) 75

Q.93. Jupiter and Saturn take 9 hours 50 minutes and 11 hours 40 minutes respectively to complete one rotation. What is the ratio of their rotation times in the lowest form?

Matriculation Level 26/07/2025(Shift - 1)

(a) 59 : 70 (b) 49 : 58
(c) 59 : 69 (d) 50 : 68

Q.94. Rahim earns ₹16,000 per month, and Ami earns ₹20,000 per annum. If both spend ₹12,000 per month, what is the ratio of their savings?

Matriculation Level 28/07/2025(Shift - 1)

(a) 1 : 2 (b) 2 : 3 (c) 1 : 3 (d) 4 : 5

Q.95. What is the mean proportional between 9 and 25 ?

Matriculation Level 29/07/2025(Shift - 1)

(a) 12 (b) 13 (c) 15 (d) 20

Q.96. If the ratio is $21 : 3 = 7 : y$. Then find the y.

Matriculation Level 30/07/2025(Shift - 1)

(a) 7 (b) 4 (c) 5 (d) 1

Q.97. If 10, 15, x are in proportion. Find the value of x.

Matriculation Level 31/07/2025(Shift - 1)

(a) $\frac{45}{2}$ (b) $\frac{55}{2}$ (c) $\frac{15}{2}$ (d) $\frac{30}{4}$

Q.98. On a certain assembly line the rejection rate for Maruti Alto production was 5 percent for Maruti Swift production 9 percent and the two cars combined 8 percent. What was the ratio of Maruti Swift production to Maruti Alto production?

Matriculation Level 31/07/2025(Shift - 1)

(a) 3 : 2 (b) 3 : 1 (c) 5 : 3 (d) 7 : 2

Q.99. In a school's sports department, the ratio of the number of boys to girls is 7 : 4. If there are 196 girls, the total number of students in the sport's department is:

Matriculation Level 01/08/2025(Shift - 1)

(a) 420 (b) 539 (c) 512 (d) 490



Q.100. Aman collection contains Spain, Japan, and France stamps. If the ratio of Spain to Japan stamps is 3 to 4 and the ratio of Japan to France stamps is 6 to 5, what is the ratio of Spain to France stamps?

Matriculation Level 01/08/2025(Shift - 1)
(a) 9 : 10 (b) 5 : 4 (c) 3 : 5 (d) 18 : 5

Q.101. A sum of Rs. 630 is to be divided among A, B and C such that 3 times A's share, 6 times B's share and 5 times C's share are all equal. The share of C is:

Matriculation Level 01/08/2025(Shift - 2)
(a) Rs. 150 (b) Rs. 180
(c) Rs. 200 (d) Rs. 225

Q.102. A recipe uses flour and cocoa powder in the ratio 5 : 2. If you have 500 grams of flour, how much cocoa powder do you need to keep the same ratio?

Matriculation Level 01/08/2025(Shift - 3)
(a) 150 (b) 180 (c) 200 (d) 250

Q.103. The present ratio of students to teachers at a certain college is 40 : 1. If the student enrollment were to increase by 60 students and the number of teachers were to increase by 4, the ratio of the teachers would then be 30 : 1. What is the present number of teachers?

Matriculation Level 01/08/2025(Shift - 3)
(a) 6 (b) 5 (c) 8 (d) 10

Q.104. If the ratio is $36 : 8 = 9x : 16$. Then find the x.

Matriculation Level 02/08/2025(Shift - 1)
(a) 6 (b) 7 (c) 8 (d) 9

Q.105. Two numbers are in ratio 3 : 7. If the sum is 80, find the numbers.

Higher Secondary 26/07/2025 (Shift - 2)
(a) 24:56 (b) 27:53 (c) 21:56 (d) 38:42

Q.106. In a factory, the ratio of skilled to unskilled workers was 8 : 5. After a recruitment drive, the number of skilled workers increased by 15%, while the number of unskilled workers increased by 20%. What is the new ratio of skilled to unskilled workers?

Higher Secondary 28/07/2025 (Shift - 2)
(a) 21 : 12 (b) 23 : 15 (c) 8 : 5 (d) 21 : 25

Q.107. Find the fourth proportional to 32, 48, and 64.

Higher Secondary 30/07/2025 (Shift - 2)
(a) 84 (b) 96 (c) 48 (d) 60

Q.108. The ratio of the number of boys to the number of girls in a class is 10 : 13. If there are 115 students in the class, how many boys are there?

Higher Secondary 30/07/2025 (Shift - 2)
(a) 50 (b) 75 (c) 85 (d) 60

Q.109. The ratio of the ages of two friends is 5 : 6. After 15 years, the ratio becomes 8 : 9. What are their present ages?

Higher Secondary 02/08/2025 (Shift - 2)
(a) 20 and 30 (b) 52 and 58
(c) 25 and 30 (d) 24 and 28

Q.110. Three people P, Q, and R, have incomes that are in the ratio 13 : 14 : 15. If their total earnings are Rs. 2,31,000, how much money does Q earn?

Higher Secondary 02/08/2025 (Shift - 2)
(a) Rs. 91,000 (b) Rs. 93,500
(c) Rs. 1,04,500 (d) Rs. 77,000

eduquity-based pattern (ebp) Questions

Q.111. If a man's age is four times the sum of the ages of his three children, and after 6 years his age will be twice the sum of the ages of his three children, then what is the man's present age?

(a) 60 years (b) 56 years
(c) 45 years (d) 30 years

Q.112. In a student council election, there were three contestants. All the voters participated in the voting and 10% of votes were declared invalid. The contestants got the votes in the ratio 8 : 7 : 5. If the second candidate in the list got 4725 votes, the total number of valid votes is:

(a) 10,000 (b) 11,500 (c) 13,500 (d) 9,000

Q.113. 141 is divided into two parts in such a way that the one-eighth part of the first and one-ninth part of the second are in the ratio 5 : 6. Find the first part.

(a) 36 (b) 72 (c) 48 (d) 60

Q.114. The salaries of A and B are in a ratio of 2 : 3. Their savings are equal. If

A's expenditure is $\frac{1}{2}$ of the salary, then what per cent of income does B spends?

(a) 66.67% (b) 78.5% (c) 50% (d) 75%

Q.115. 360 toffees are distributed among Pushkal, Quadir and Rustam in the ratio 2 : 3 : 4. After distribution, Quadir gives 20% of his toffees to Pushkal and Rustam gives 25% of his toffees to Pushkal. What is the new ratio of toffees with Pushkal, Quadir and Rustam?

(a) 5 : 4 : 6 (b) 7 : 3 : 5
(c) 6 : 4 : 5 (d) 3 : 5 : 7

Q.116. Due to a fault in the voting machine, the invalid votes are counted in votes of the candidate B, who gets 5 times the invalid votes. A gets the votes

8 times the invalid votes and wins the election by 28. What is the total number of votes polled?

(a) 168 (b) 184 (c) 196 (d) 176

Q.117. The mean proportional between the numbers p and q is 8. Which of the following pairs of numbers can be the values of p and q?

(a) 12 and 3 (b) 16 and 4
(c) 12 and 16 (d) 10 and 6

Q.118. The ratio of the incomes of Rama and Ganga is 4 : 3 and the ratio of their expenditures is 3 : 2. If each person saves ₹2,700, then find their expenditures, respectively.

(a) ₹8,100 and ₹5,000
(b) ₹8,000 and ₹5,000
(c) ₹8,100 and ₹5,400
(d) ₹8,000 and ₹5,400

Q.119. Shankar comes to buy a T-shirt in a showroom. He purchases a shirt which costs ₹1,020 after a discount. He pays his bill in the form of 50 paise, 20 paise and 10 paise coins in the ratio 2 : 3 : 4 (assume for the sake of the problem). What is the number of 10 paise coins he used to pay his bill?

(a) 1530 (b) 510 (c) 1020 (d) 2040

Q.120. What should be added to each of the four numbers 5, 19, 7 and 25 to make them proportional?

(a) 2 (b) 3 (c) 1 (d) 4

Q.121. If $49 : x :: x : 81$ and $64 : y :: y : 169$, where x and y are both natural numbers, then find the value of $2x + 3y$.

(a) 312 (b) 126 (c) 348 (d) 438

Q.122. When x is added to each of 2, 3, 30 and 35, then the numbers obtained in this order are in proportion. What is the second largest part in the proportion?

(a) 40 (b) 30 (c) 35 (d) 45

Q.123. The quantity of milk ordered on the first, second, third and fourth day of a four day ceremony is in the proportion of 30 litres, 24 litres, 35 litres and 'x' litres, respectively. Find the quantity of milk (in litres) ordered on the fourth day.

(a) 30 (b) 32 (c) 28 (d) 26

Q.124. The average marks (out of 100) of boys and girls in an examination are 75 and 80, respectively. If the average marks of all the students in that examination are 78. Find the ratio of the number of boys to the number of girls.

(a) 2 : 3 (b) 3 : 4 (c) 1 : 3 (d) 1 : 2

Q.125. At present, A is thrice as old as B. C is 5 years elder than B. The sum of the



ages of A, B, C is 75 years. The ratio of the age of B five years ago to the age of A three years from now is:

- (a) 2 : 3 (b) 1 : 4 (c) 4 : 7 (d) 1 : 5

Q.126. If x varies inversely as y and $x = 2$ when $y = 6$, then the value of y when $x = 3$ is:

- (a) 2 (b) 3 (c) 4 (d) 6

Q.127. The ages of two friends, Ram and Ravi, differ by 7 years. Ram's mother Mayadevi is three times as old as Ram, and Ravi is four times as old as his brother Soham. The ages of Mayadevi and Soham differ by 65 years. Find the age of Mayadevi (in years).

- (a) 69 (b) 36 (c) 85 (d) 45

Q.128. What is the value of D, if $A : C = B : D$ in the given table?

A	B	C	D
25	40	30	?

- (a) 56 (b) 48 (c) 45 (d) 42

Q.129. A money bag contains one rupee, 5 rupee and 10 rupee coins in the ratio of 3 : 5 : 7 respectively. If the total amount in the bag is ₹980, then find the number of coins of ₹10.

- (a) 69 (b) 70 (c) 71 (d) 68

Q.130. A sum of money is divided among P, Q, R and S in the proportion of 4 : 9 : 2 : 3. If Q gets ₹560 more than R, then the share of S is:

- (a) ₹140 (b) ₹340 (c) ₹240 (d) ₹440

Q.131. The incomes of P, Q and R are in the ratio 10 : 12 : 9 and their expenditures are in the ratio 12 : 15 : 8. If Q saves 25% of his income, then what is the ratio of the savings of P, Q and R?

- (a) 15 : 14 : 21 (b) 14 : 15 : 21
(c) 21 : 15 : 14 (d) 21 : 14 : 15

Q.132. Raj divides ₹1,200 in the ratio 2 : 1 : 3 among three of his friends. The amount equal to the sum of three times the largest share and two times the smallest share is:

- (a) ₹2,400 (b) ₹1,800 (c) ₹2,200 (d) ₹2,000

Q.133. The ratio between male and female

members in a club is 2 : 3. If the number of male members is increased by 200, the ratio becomes 5 : 6. How many female members are there in the club?

- (a) 1200 (b) 900 (c) 1400 (d) 1000

Q.134. Find the mean proportional of 12.96 and 0.16.

- (a) 1.21 (b) 1.96 (c) 1.44 (d) 1.69

Q.135. In a collection of rare coins, there

is one gold coin for every four non-gold coins. If 20 more gold coins are added to the collection, the ratio of the number of gold coins to that of non-gold coins will be 2:3. The total number of coins in the collection will now become _____.

- (a) 80 (b) 60 (c) 100 (d) 48

Q.136. In a company, the ratio of men and women is 5 : 3, respectively. When 150 additional women are employed in the company, the ratio becomes 5 : 4. How many men are employed in the company?

- (a) 850 (b) 750 (c) 550 (d) 950

Q.137. There are y yellow marbles and b blue marbles in a box. 18% of the yellow marbles were added to 24% of the blue marbles, and the total was 30. Three times the number of yellow marbles exceeds two times the number of blue marbles by 200. What is the ratio of y^2 to b^2 ?

- (a) 6 : 5 (b) 4 : 1 (c) 2 : 1 (d) 5 : 2

Q.138 The mean proportional of $4a^2$ and x is $\frac{1}{5a}$, where a is a positive number, then the value of x is _____

- (a) $\frac{1}{100a^4}$ (b) $\frac{1}{100a^2}$ (c) $\frac{1}{25a^2}$ (d) $\frac{1}{25a^4}$

Q.139. The cost of a diamond varies directly as the square of its weight. Once, this diamond broke into four pieces with weights in the ratio of 4 : 3 : 2 : 1. When the pieces were sold, the merchant got ₹63,000 less than that of one piece. Find the original price of the diamond.

- (a) ₹86,000 (b) ₹80,000
(c) ₹90,000 (d) ₹75,000

Q.140. The ratio of boys to girls initial in a college is 5 : 6. If 40 boys leave the college and 50 girls join the college, the ratio will become 8 : 11. Find the number of girls (initial) in the college.

- (a) 740 (b) 860 (c) 720 (d) 640

Q.141. 30 is the mean proportional of 18 and A . Find the value of A .

- (a) 60 (b) 40 (c) 45 (d) 50

Q.142. The salaries of P and Q together amount to ₹9,60,000. P and Q save 64% and 44%, respectively of their individual salaries. If P's expenditure is 1.5 times Q's expenditure, then what is the ratio of P's salary to Q's salary?

- (a) 2 : 3 (b) 7 : 3 (c) 3 : 7 (d) 3 : 2

Q.143. The ratio of circumference and diameter of a circle is 3 : 5. If the circumference is 6 cm, then the radius of the circle is:

- (a) 10 cm (b) 15 cm (c) 5 cm (d) 1.8 cm

Q.144. If $a : b = c : d = e : f = 5 : 7$, then what is the ratio $(3a + 5c + 11e) : (3b + 5d + 11f)$?

- (a) 7 : 11 (b) 3 : 7 (c) 5 : 7 (d) 11 : 7

Q.145. The monthly incomes of A, B and C are in the ratio 2 : 9 : 3 and their expenses are in the ratio 3 : 9 : 5. If A's saving is half of his total income, then savings of A, B and C are, respectively, in the ratio of:

- (a) 3 : 9 : 2 (b) 2 : 5 : 2
(c) 3 : 18 : 4 (d) 1 : 2 : 1

Q.146. Radhika requested the cashier of a bank to get her coins in lieu of her cheque worth ₹1,845. Cashier gave her a packet containing ₹5, ₹10 and ₹20 coins in the ratio of 3 : 5 : 7. What is the total worth of the smallest coin received by her?

- (a) ₹135 (b) ₹125 (c) ₹145 (d) ₹105

Q.147. The age of the older of two boys is thrice that of the younger. 9 years ago it was five times that of the younger. Find the present age of each.

- (a) 15 years, 45 years
(b) 18 years, 54 years
(c) 17 years, 51 years
(d) 12 years, 36 years

Q.148. P pays Q a sum of ₹150 using coins of ₹2, ₹5 and ₹10. He uses a total of 50 coins. If the ratio of ₹2 and ₹5 coins used is 5 : 2, then how many coins of ₹10 are used in the payment?

- (a) 2 (b) 1 (c) 4 (d) 5

Q.149. In what ratio must a grocer mix two varieties of pulses costing ₹60 and ₹75 per kg, respectively, so as to get a mixture worth ₹65 per kg?

- (a) 3 : 4 (b) 2 : 1 (c) 2 : 3 (d) 1 : 2

Q.150. M varies inversely as $(N^2 + 3)$. If $M = 3$ when $N = 3$, then what will be the value of M when $N^2 = 69$?

- (a) 1 (b) $\frac{1}{4}$ (c) $\sqrt{69}$ (d) $\frac{1}{2}$

Q.151. The difference between two numbers is 48. The ratio of these two numbers is 7 : 3.

What is the sum of the two numbers?

- (a) 110 (b) 100 (c) 90 (d) 120

Q.152. Divide some money in the ratio Ravi, Reeta and Rahul that 5 (Part of Ravi) = 3 (Part of Reeta) = 11 (Part of Rahul). The money ratio of Ravi : Reeta : Rahul is equal to :

- (a) 11 : 5 : 3 (b) 11 : 33 : 15
(c) 5 : 11 : 3 (d) 33 : 55 : 15



Q.153. Among the ratios 3 : 4, 5 : 8, 1 : 2, 6 : 9 which is the greatest?

- (a) 5 : 8 (b) 6 : 9 (c) 1 : 2 (d) 3 : 4

Q.154. If $a : b = 3 : 7$ and $b : c = 4 : 9$, then find $a : b : c$.

- (a) 12 : 28 : 63 (b) 3 : 28 : 63
(c) 12 : 28 : 9 (d) 3 : 28 : 9

Q.155. Which is the greatest ratio among the following?

$$\frac{8}{11}, \frac{4}{5}, \frac{1}{2} \text{ and } \frac{1}{11}$$

- (a) $\frac{4}{5}$ (b) $\frac{8}{11}$ (c) $\frac{1}{2}$ (d) $\frac{1}{11}$

Q.156. When 'X' is added to each of the numbers 7, 11, 18 and 23, then the number so obtained are in proportion. What is the mean proportional between the $(x - 1)$ and $(2x - 10)$?

- (a) 78 (b) 98 (c) 48 (d) 68

Q.157. Convert 28% into the form of ratio.

- (a) 25 : 7 (b) 7 : 25 (c) 7 : 18 (d) 18 : 7

Q.158. A box filled with gift articles weighs 25 kg. If the weight of the box and the gift articles, respectively, are in the ratio 1 : 9, then the weight of the articles (in grams) is:

- (a) 22500 (b) 23500
(c) 21500 (d) 24500

Q.159. Find the third proportion to 36 and 48.

- (a) 36 (b) 54 (c) 48 (d) 64

Q.160. If $a : b = 3 : 4$, $b : c = 2 : 3$, and $c : d = 5 : 6$, then $a : b : c : d$ is:

- (a) 10 : 20 : 30 : 36 (b) 15 : 25 : 30 : 36
(c) 15 : 20 : 30 : 36 (d) 15 : 20 : 35 : 36

Q.161. If $\frac{1}{M} : \frac{1}{N} : \frac{1}{O} = 3 : 4 : 5$, then $M : N : O$ is equal to:

- (a) 12 : 15 : 20 (b) 20 : 15 : 12
(c) 20 : 12 : 15 (d) 15 : 20 : 12

Q.162. The mean proportion of 0.14 and 2.24 is:

- (a) 0.56 (b) - 2.39 (c) 1.12 (d) 1.02

Q.163. Find the fourth proportion to 1.25, 5.75 and 3.5

- (a) 4.5 (b) 16.1 (c) 12.5 (d) 6.5

Q.164. Two numbers are in the ratio 2 : 3. Their sum is 135. Find the numbers.

- (a) 54, 81 (b) 45, 90 (c) 50, 85 (d) 60, 75

Q.165. Find the mean proportional between 25 and 81.

- (a) 12 (b) 36 (c) 81 (d) 45

Q.166. Two numbers are, respectively, 30% and 60% more than a third number.

The ratio of the two numbers is:

- (a) 13 : 16 (b) 12 : 13
(c) 11 : 15 (d) 13 : 17

Q.167. The strength of a school is 640, with the number of boys being 400. Find the ratio of girls to boys.

- (a) 5 : 6 (b) 3 : 5 (c) 4 : 5 (d) 8 : 5

Q.168. Raju and Ramu have more land than Sonu by 15% and 25%, respectively. What is the ratio of land owned by Raju and Ramu?

- (a) 56 : 25 (b) 21 : 23
(c) 23 : 25 (d) 25 : 23

Q.169. If $x : y = 5 : 6$, then $2x + 3y : 3x + 5y$ is:

- (a) 18 : 35 (b) 28 : 45
(c) 18 : 45 (d) 28 : 48

Q.170. Which of the following ratios is greater?

- (a) 7 : 4 (b) 8 : 3 (c) 2 : 3 (d) 1 : 5

Q.171. The ratio of red to green sweets in a bag is 3 : 4. How many red sweets are there in the bag if the number of green sweets is 120?

- (a) 80 (b) 100 (c) 70 (d) 90

Q.172. Find the mean proportion to 0.72 and 2.85 (round up to one decimal place).

- (a) 1.4 (b) 0.7 (c) 2.8 (d) 0.4

Q.173. What should be added to each term of the ratio 5 : 11 so that the ratio becomes 3 : 5?

- (a) 5 (b) 4 (c) 2 (d) 6

Q.174. Two numbers are in the ratio of 3 : 7. If 11 is subtracted from each, the ratio between the numbers becomes 7 : 20. Find the smaller number.

- (a) 39 (b) 21 (c) 36 (d) 27

Q.175. If 40% of $A = 60\%$ of $B = \frac{1}{6}$ of C , then find the ratio of $A : B : C$.

- (a) 15:10:18 (b) 36:10:15
(c) 15:10:36 (d) 10:15:36

Q.176. The ratio of marks of A and B is 3 : 4 and the ratio of marks of B and C is 4 : 5. If the difference in the marks of C and A is 30, then how much marks does B get?

- (a) 52 (b) 64 (c) 60 (d) 56

Q.177. A sum of ₹17,600 was divided between Jeevan, Praveen and Kanan, such that the ratio of the sums received by Jeevan and Praveen was 5 : 8, while the ratio of the sums received by Praveen and Kanan was 3 : 2. How much money did Jeevan receive as his share?

- (a) ₹4,900 (b) ₹4,750
(c) ₹4,850 (d) ₹4,800

Q.178. Find the natural number obtained when the number 13 is subtracted from the mean proportional of 75 and 12.

- (a) 14 (b) 30 (c) 27 (d) 17

Q.179. If four distinct positive numbers, a, b, c and d, in the order given are in proportion, then which of the following options is NOT correct?

- (a) b, a, d and c, in the order given are in proportion.
(b) c, d, b and a, in the order given are in proportion.
(c) d, c, b and a, in the order given are in proportion.
(d) a, c, b and d, in the order given are in proportion.

Q.180. Let 'a' be a positive number and $b = \frac{4}{a}$. What is the mean proportional of a and b

- (a) 4 (b) 3 (c) 2 (d) a

Q.181. What is the ratio between S and T if S is 75% of T?

- (a) 4 : 3 (b) 2 : 3 (c) 75 : 1 (d) 3 : 4

Q.182. In a school, the total number of boys is 20% more than the total number of girls. What is the ratio of the number of boys to that of girls in the school?

- (a) 6 : 5 (b) 5 : 6 (c) 5 : 3 (d) 4 : 5

Q.183. 42% of Ranita's weekly income is equal to 56% of Bhaskar's weekly income. If Bhaskar's weekly income was raised by ₹ 200, while Ranita's weekly income did not change, the ratio of the weekly incomes of Ranita and Bhaskar, respectively, would have been 5 : 4. What is Ranita's weekly income (in ₹)?

- (a) 4200 (b) 3600 (c) 3000 (d) 4000

Q.184. Swati earns a sum of ₹40,000, out of which she saves 25%. She spends the remaining amount on food and education in the ratio of 3 : 2. What is the percentage of expenditure incurred on education to her total income?

- (a) 30% (b) 20% (c) 60% (d) 40%

Q.185. In an educational institution, the ratio of the numbers of students in middle school and in high school, respectively, in a given year was given as 4 : 3. If there were 324 students in middle school in that year in that institution, what was the number of students in high school in the same year in that institution?

- (a) 243 (b) 240 (c) 249 (d) 246

Q.186. If class A has 11 boys and 18



girls, class B has 13 boys and 17 girls, and class C has 20 boys and 9 girls, which class has the biggest ratio(s) of boys to girls?

- (a) Class A (b) Both class A and class B
(c) Class C (d) Class B

Q.187. A recipe is made using cheese and butter in the ratio 4 : 3. If Arohi uses 8 bowls of cheese, how many bowls of butter should she use?

- (a) 4 (b) 6 (c) 8 (d) 3

Q.188. The ratio of the salaries of S to F is 2 : 3 and the ratio of their savings is 3 : 4. What is the ratio of their expenditures if their total savings are equal to the salary of S?

- (a) 8 : 15 (b) 7 : 15 (c) 8 : 13 (d) 7 : 12

Q.189. S varies directly as $(R + 7)$, and $S = 42$ when $R = 17$. What is the value of S when $R = 29$?

- (a) 51 (b) 63 (c) 58 (d) 66

Q.190. The price of a television set and a washing machine are in the ratio of 8 : 7. If the cost of the television set is ₹4,800 more than the cost of the washing machine, then find the price of the washing machine.

- (a) ₹33,000 (b) ₹33,500
(c) ₹33,600 (d) ₹33,700

Q.191. Pooja, Chameli and Anju have a total of ₹450 in their purses. Pooja has ₹50 more than Chameli. Anju has two times what Pooja has. How much do Pooja, Chameli and Anju, respectively, have in their purses?

- (a) ₹150, ₹75 and ₹225
(b) ₹100, ₹50 and ₹300
(c) ₹125, ₹75 and ₹250
(d) ₹125, ₹50 and ₹275

Q.192. Two numbers are in the ratio 4 : 5. If 3 is added to the smaller and 7 is subtracted from the larger number, they are in the ratio 1 : 1. Find the numbers.

- (a) 60, 75 (b) 40, 50 (c) 55, 44 (d) 20, 25

Q.193. What is the mean proportional of $45x^4$ and $5y^2$?

- (a) $5x^2y$ (b) $25x^4y$ (c) $15x^2y$ (d) $45x^2y^2$

Q.194. The ages of A and B are in the ratio 4 : 7, respectively. After 8 years, the ratio of their ages will be 2 : 3. What is the sum of the ages of A and B (in years)?

- (a) 44 (b) 36 (c) 38 (d) 42

Q.195. Which number should be added to 3, 11, 4 and 14 to make them in proportion?

- (a) 3 (b) 2 (c) 4 (d) 1

Q.196. Two numbers are in the ratio 4 : 5. If 17 is subtracted from each, the new numbers are in the ratio 11 : 14. If 20 is added to each of the original numbers, then the ratio becomes:

- (a) 224 : 275 (b) 275 : 241
(c) 241 : 275 (d) 224 : 241

Q.197. A bag contains a total of 180 coins in the denominations of ₹5 and ₹1. Find the number of ₹5 coins in the bag, if the total value of the coins is ₹500.

- (a) 60 (b) 120 (c) 80 (d) 100

Q.198. Find the third proportional of 9, 7, 7, 14.

- (a) 21 (b) 16 (c) 24 (d) 18

Q.199. Find the ratio between the fourth proportional of 42, 56, 63 and the fourth proportional of 189, 273, and 153.

- (a) 33 : 221 (b) 217 : 42
(c) 33 : 84 (d) 84 : 221

Q.200. Rishi and Sarita have ₹4,760. If $\frac{9}{16}$ of Rishi's amount is equal to $\frac{3}{4}$ of Sarita's amount, how much does Sarita have?

- (a) ₹3,570 (b) ₹1,190
(c) ₹2,720 (d) ₹2,040

Q.201. A sum of money was divided between Cyrus, Rohan and Mishti. The ratio of the sums received by Cyrus and Rohan was 3 : 2, while the ratio of the sums received by Rohan and Mishti was 7 : 12. If the sum received by Cyrus was ₹1,800 less than the sum received by Mishti, how many rupees did Rohan receive as his share?

- (a) ₹9,240 (b) ₹7,560
(c) ₹8,400 (d) ₹6,300

Q.202. The third proportional to $(5^2 - 3^2)$ and $(5 - 3)$ is:

- (a) $\frac{1}{3}$ (b) $\frac{1}{8}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

Q.203. Four distinct positive numbers, 'a', 'b', 'c' and 'd', in the order given, are in proportion. 'b' is 35 more than 'a' and 'd' is 60 more than 'c'. The product of 'a' and 'c' is 5376. What is the sum of 'a', 'b', 'c' and 'd'?

- (a) 398 (b) 400 (c) 399 (d) 397

Q.204. A bag contains rupee, 50-paise and 25-paise coins in the ratio 3 : 4 : 6. If the total amount is ₹143, the number of 50-paise coins is:

- (a) 132 (b) 66 (c) 90 (d) 88

Q.205. The sum of the four numbers A, B, C and D is 875. If the ratio of A to B is 1 : 2, the ratio of B to C is 3 : 1 and the ratio of C to D is 2 : 3, find the value of C.

- (a) 120 (b) 130 (c) 125 (d) 135

Q.206. The ratio of the expenditure to the savings of Richa is 2 : 1. If her income and expenditure increase by 20% and 15%, respectively, then find the percentage increase in her savings.

- (a) 35% (b) 30% (c) 20% (d) 25%

Q.207. The ratio of prices of Coffee and Tea is 3 : 4 and the ratio of their quantities consumed by a family is 4 : 7. Find the ratio of expenditures on Coffee and Tea.

- (a) 3 : 7 (b) 7 : 3 (c) 7 : 4 (d) 4 : 7

Q.208. ₹7,786 is divided among A, B and C in such a manner that the ratio of the amount of A to that of B is 3 : 4 and the ratio of the amount of B to that of C is 12 : 13. The amount of money received by B is:

- (a) ₹7,138 (b) ₹2,977
(c) ₹2,748 (d) ₹5,038

Q.209. The areas of the two triangles are in the ratio 4 : 3 and their heights are in the ratio 6 : 5. Find the ratio of their bases.

- (a) 5 : 6 (b) 10 : 9 (c) 9 : 10 (d) 6 : 5

Q.210. If $\frac{x}{4y} = \frac{3}{4}$, then the value of $\frac{2x + 3y}{x - 2y}$ is

- (a) 8 (b) 6 (c) 9 (d) 7

Q.211. P and Q are inversely proportional to each other and P is 10, when Q is 6. What will be the value of Q when P is 15?

- (a) 9 (b) 4 (c) 10 (d) 5

Q.212. The mean proportion of $(3 + 2\sqrt{2})^2$ and $(3 - 2\sqrt{2})^2$ is:

- (a) $3 + 2\sqrt{2}$ (b) 5 (c) $2\sqrt{2}$ (d) 1

Q.213. Profit of ₹29,120 is divided among A, B and C so that A's share is $\frac{1}{2}$ times B's and B's share is $1\frac{1}{5}$ times C. Find B's share.

- (a) ₹6,940 (b) ₹7,820 (c) ₹6,720 (d) ₹7,140

Q.214. Rupees 66,000 is to be divided amongst Asha, Bhupinder and Gargi in such a way that ratio of their shares is to be: Asha : Bhupinder = 9 : 5 and Bhupinder : Gargi = 2 : 1. How much should Asha get?

- (a) ₹38,000 (b) ₹36,000
(c) ₹32,000 (d) ₹34,000

Q.215. A certain sum is to be divided between A, B, and C in the ratio 2 : 5 : 7. But by mistake, it was divided in the ratio 5 : 2 : 7. Thus B loses by ₹150. Find the



sum.

(a) ₹700 (b) ₹850 (c) ₹900 (d) ₹625

Q.216. If $p : q = 5 : 2$, then $p^2 - q^2 : p^2 + q^2 = ?$

(a) 29 : 21 (b) 4 : 3 (c) 3 : 4 (d) 21 : 29

Q.217. The marks scored by a student are directly proportional to the time invested in studies. If the student gets 60 marks when he studies 4 hours a day, then how many hours he should use in his studies daily to get 75 marks ?

(a) 4.5 (b) 5 (c) 5.5 (d) 6

Q.218. Three numbers are in the ratio 5 : 2 : 7. The sum of their squares is 3822. The biggest number is :

(a) 35 (b) 38 (c) 78 (d) 49

Q.219. Among four numbers the sum of the smallest and largest is 72 more than the sum of middle two numbers. If the numbers are in the ratio 6 : 7 : 15 : 19, then the smallest number is :

(a) 72 (b) 144 (c) 168 (d) 216

Q.220. If the income of T is 8 times his savings, then his expenditure is ___ times his savings.

(a) 6 (b) 7 (c) 8 (d) 5

Q.221. The ratio of the number of boys to the number of girls in a school of 1650 students is 6 : 5. If 124 new boys are admitted and few new girls are admitted, the ratio changes to 4 : 3. What is the ratio of the newly admitted girls to the newly admitted boys ?

(a) 3 : 4 (b) 62 : 9 (c) 4 : 3 (d) 9 : 62

Q.222. In a class of boys and girls, the number of girls is twice the number of boys. If 20 more boys join the class, the ratio of girls and boys is 1 : 1. How many total students were there in the class initially ?

(a) 60 (b) 75 (c) 45 (d) 90

Q.223. If $A : B : C = 3 : 4 : 5$, then $\sqrt{A^2 + B^2} : \sqrt{C^2 - B^2}$ is:

(a) 3 : 2 (b) 4 : 5 (c) 5 : 3 (d) 3 : 5

Q.224. The cost of 3 cups and 5 plates is ₹1,080 and the cost of 2 cups and 4 plates is ₹840. Find the ratio of the cost of a cup to the cost of a plate.

(a) 1 : 3 (b) 2 : 3 (c) 1 : 1 (d) 1 : 2

Q.225. The compound ratio of 3 : 4 and 6 : 7 is 36 : x. Find x.

(a) 64 (b) 75 (c) 56 (d) 48

Q.226. Find 2 numbers such that their mean proportional is 25 and their third proportional is 25.

(a) 5 and 5 (b) 5 and 25

(c) 25 and 125 (d) 25 and 25

Q.227. The present age of a father is twice that of his son. 8.5 years hence the ratio would be 7 : 4. Find the son's present age.

(a) 25.5 years (b) 16.5 years
(c) 20.5 years (d) 24.6 years

Q.228. The sum of the amount that Sunaina and Ranjan have is ₹23 and the difference of their amount is ₹7. The ratio of the amount that Sunaina and Ranjan have is:—

(a) 5 : 3 (b) 7 : 15 (c) 15 : 8 (d) 2 : 17

Q.229. Two numbers are in the ratio of 1 : 2. If 7 is added to both, their ratio changes to 3 : 5. The greater number is:

(a) 30 (b) 18 (c) 14 (d) 28

Q.230. The difference between 24 and the mean proportional of 7 and 63 is:

(a) 3 (b) 2 (c) 5 (d) 4

Q.231. If $a : b = 5 : 7$, then $(6a^2 - 2b^2) : (b^2 - a^2)$ will be :

(a) 12 : 5 (b) 21 : 5 (c) 13 : 6 (d) 17 : 8

Q.232. What is the fourth proportional of $3\sqrt{5}$, $5\sqrt{8}$ and $3\sqrt{10}$?

(a) $10\sqrt{5}$ (b) $40\sqrt{2}$ (c) 30 (d) 20

Q.233. The ratio of the father's age to his son's age is 5 : 3. The product of the numbers representing their ages is 960. The ratio of their ages after 6 years will be:

(a) 23:15 (b) 23:17 (c) 21:15 (d) 21:17

Q.234. Find the third proportion of P and 12, when 8, P, 16 and 18 are in proportion.

(a) 16 (b) 8 (c) 18 (d) 12

Q.235. The total number of workers in a company is 960. If the number of female workers in the company is 420, then the respective ratio of the total number of male workers to the total number of female workers in the company is:

(a) 7 : 9 (b) 16 : 9 (c) 16 : 7 (d) 9 : 7

Q.236. Given that x varies directly as y and $x = 45$ when $y = 9$. What is the value of x when $y = 50$?

(a) 225 (b) 190 (c) 175 (d) 250

Q.237. Divide ₹ 1488 among X, Y and Z. If the shares are in the ratio $X : Y = 4 : 5$, $Y : Z = 7 : 6$, then find out the share of Z.

(a) ₹ 460 (b) ₹ 450 (c) ₹ 470 (d) ₹ 480

Q.238. The ratio of the number of boys to that of the girls in a school is 11 : 15. If there are 200 more girls than boys in the

school, what is the number of boys in that school ?

(a) 550 (b) 750 (c) 506 (d) 605

Q.239. The ratio between the monthly income and the expenditure of Vaidic is 8 : 5. If his income increases by 20% and his expenditure increases by 30%, then find the percentage increase or decrease in his monthly savings.

(a) $5\frac{2}{3}\%$ increase (b) $5\frac{2}{3}\%$ decrease

(c) $3\frac{1}{3}\%$ increase (d) $3\frac{1}{3}\%$ decrease

Q.240. In a factory, 48% of the number of male workers is equal to two-third of the number of female workers. What is the ratio of the number of males to that of the females in the factory ?

(a) 25 : 12 (b) 12 : 25 (c) 18 : 25 (d) 25 : 18

Q.241. Two numbers are, respectively, 17% and 50% more than a third number. The ratio of the two numbers is:

(a) 27 : 25 (b) 39 : 50

(c) 19 : 11 (d) 29 : 25

Q.242. If $a : b :: b : c$, and $b = 96$, then which of the following can be a possible pair of values of a and c ?

(a) $a = 24$; $c = 374$ (b) $a = 32$; $c = 288$

(c) $a = 48$; $c = 168$ (d) $a = 16$; $c = 586$

Q.243. Three numbers are in the ratio $\frac{4}{5} : \frac{5}{6} : \frac{9}{10}$. The difference between the smallest and the greatest numbers is 12. Find the number which is neither the smallest nor the greatest.

(a) 96 (b) 108 (c) 100 (d) 104

Q.244. A pole is 9030 cm long. It is colored saffron, white and green in the ratio of 6 : 7 : 8. Find the length of the green part of the pole.

(a) 3440cm (b) 3010cm

(c) 3480cm (d) 2580cm

Q.245. If a 7-storey building has a 28 m long shadow, the number of storeys of the building whose shadow is 48 m long is:

(a) 14 (b) 24 (c) 16 (d) 12

Q.246. What smallest positive number should be subtracted from 2 and 3 so that 4 is the third proportion to them?

(a) 1 (b) 2 (c) 4 (d) 3

Q.247. If the third proportional of $3x^2$ and $4xy$ is 48, then find the positive value of y.

(a) 3 (b) 5 (c) 2 (d) 4

Q.248. The ratio of cheese to sauce for a single pizza is 1 cup to $\frac{1}{2}$ cup. If Rama



used 15 cups of sauce to make a number of pizzas, then how many cups of cheese did she use on those pizzas ?

- (a) 20 cups (b) 10 cups
(c) 40 cups (d) 30 cups

Q.249. Out of the total income, a family saves 15% and the rest of the amount is spent on three types of expenditure - health, education and food in the ratio of 1 : 2 : 2. The income is Rs.300 less than thrice the expenditure incurred on education. What is the difference between the savings and the expenditure incurred on health?

- (a) Rs.300 (b) Rs.350
(c) Rs.2,250 (d) Rs.500

Q.250. Let M denote the mean proportional between 16 and 9 and let N denote the third proportional of 9 and 15. Find the value of $5M + 3N$.

- (a) 300 (b) 125 (c) 135 (d) 225

Q.251. A company is manufacturing three different products - Glass, Plate and Bowl. The number of Plates is double the number of Glasses. The number of Glasses manufactured is twice the number of Bowls. What is the difference between the number of Plates and the number of Bowls if the total units manufactured by the company is 2380?

- (a) 1360 (b) 680 (c) 1120 (d) 1020

Q.252. Two numbers are in the ratio of 7 : 6. If the difference between their squares is 637, then the numbers are

- (a) 49 and 42 (b) 49 and 12
(c) 43 and 36 (d) 14 and 12

Q.253. The students in three batches of a dance class are in ratio 2 : 3 : 5. If 20 students increase in each batch the ratio changes to 4 : 5 : 7. Find the total number of students in the three batches before the increase.

- (a) 120 (b) 100 (c) 150 (d) 80

Q.254. In an examination, the ratio of practical to theory marks is 3 : 7. If 80% of the marks are scored in the practical and 60% of marks are scored in the theory, what is the ratio of marks scored in the theory and the practical ?

- (a) 6 : 7 (b) 5 : 7 (c) 7 : 4 (d) 3 : 7

Q.255. If $a : b : c = 4 : 7 : 9$ and $b : c : d = 28 : 36 : 21$, then $a : b : c : d$ is:

- (a) 12 : 21 : 27 : 17 (b) 16 : 28 : 36 : 21
(c) 8 : 14 : 18 : 21 (d) 4 : 7 : 9 : 11

Q.256. The ratio of the number of boys to that of girls in a village is 5 : 3. If 40% of the boys and 60% of the girls appeared in an examination, then the ratio of the

number of boys and girls who appeared in the examination to the number who did not appear in the same examination, is :-

- (a) 17 : 21 (b) 19 : 21
(c) 17 : 27 (d) 19 : 23

Q.257. The radii of two cylinders are in the ratio 2 : 1 and their heights are in the ratio 5 : 2. The ratio of their volumes is:

- (a) 20 : 1 (b) 15 : 1 (c) 10 : 1 (d) 5 : 1

Q.258. The ratio of two numbers is 5 : 9. If 6 is added to the greater number and 6 is subtracted from the smaller number, then the greater number becomes 6 times the smaller number. What are the numbers ?

- (a) 15 and 27 (b) 10 and 18
(c) 25 and 45 (d) 20 and 36

Q.259. If $\frac{J}{4} = \frac{K}{5} = \frac{L}{3}$, then what is the value of $(L + J)^2 : (J + K)^2 : (K + L)^2$?

- (a) 25 : 16 : 9 (b) 64 : 25 : 81
(c) 49 : 81 : 62 (d) 49 : 81 : 64

Q.260. In three positive numbers, the ratio of first and second is 3 : 4 and the ratio of second and third is 3 : 5. The product of first and third is 4500. What is the sum of the three numbers?

- (a) 282 (b) 235 (c) 198 (d) 205

Q.261. When a sum of money was equally distributed among 50 children, each child received Rs. 80. If the same amount is equally distributed among another group of children such that each child get Rs. 50, then what is the number of children in the second group?

- (a) 50 (b) 80 (c) 60 (d) 40

Q.262. If $(x + y) : (y + z) : (z + x) = 3 : 5 : 7$ and $(x + y + z) = 45$, then what is the value of z ?

- (a) 18 (b) 30 (c) 27 (d) 20

Q.263. A sum of Rs. 120 is divided among p, q and r in such a way that p gets Rs. 9 more than q and q gets Rs. 6 more than r. What is the ratio of shares of p, q and r respectively ?

- (a) 18 : 15 : 13 (b) 15 : 12 : 10
(c) 16 : 13 : 11 (d) 11 : 8 : 10

Q.264. The expenditure and savings of a person are in a ratio of 4 : 3. His savings increase by $\frac{1}{5}$ of his income and income remains the same. What is the new ratio of his expenditure and savings?

- (a) 12 : 17 (b) 14 : 17
(c) 13 : 22 (d) 15 : 13

Q.265. Weight of Ayush and Abhishek

are in the ratio of 8 : 5. Abhishek's weight increases by 40 percent and the total weight of Ayush and Abhishek both increases by 60 percent. If the total weight becomes 104 kg, then what is the weight of Ayush after the increment ?

- (a) 60 kg (b) 75 kg (c) 69 kg (d) 72 kg

Q.266. The ratio of income of Jitendra, and Narendra is 4 : 7 and the ratio of their expenditure is 6 : 5 respectively. If Jitendra saves Rs.400 out of Rs.2,800 income, then find the savings of Narendra.

- (a) Rs. 2,900 (b) Rs. 2,000
(c) Rs. 3,900 (d) Rs. 4,900

Q.267. The ratio of incomes of P and Q is 1 : 2. Ratio of income of Q and R is 3 : 2. If one-third of P's income is Rs. 4400 less than the half of P's income, then what is Q's income?

- (a) Rs. 56600 (b) Rs. 52800
(c) Rs. 46600 (d) Rs. 41200

Q.268. If the difference between the two numbers is 20 percent of their sum, then what is the ratio of larger number and smaller number ?

- (a) 6 : 5 (b) 5 : 4 (c) 4 : 3 (d) 3 : 2

Q.269. The ratio of incomes of C and D is 3 : 4 respectively. The sum of their expenditure is Rs. 12000. Savings of C is the same as that of the expenditure of D. What is the sum of their savings?

- (a) Rs. 25000 (b) Rs. 15000
(c) Rs. 20000 (d) Rs. 16000

Q.270. The sum of the present ages of a father and his son is 94 years. 8 years ago their respective ages were in the ratio of 2 : 1. After 10 years what will be the ratio of ages of father and son ?

- (a) 35 : 22 (b) 14 : 13 (c) 25 : 18 (d) 16 : 15

Q.271. Rs. 43,120 is divided among P, Q and R such that P's share is $\frac{4}{7}$ of total share of Q and R together and Q's share is $\frac{2}{5}$ of total share of P and R together.

What is the share of R ?

- (a) Rs.15120 (b) Rs.14220
(c) Rs.16400 (d) Rs.18050

Q.272. The ratio of the incomes of two employees is 7 : 4, and the ratio of their expenditures is 3 : 1. If each of them manages to save ₹4,800 per month, find the sum of their monthly incomes (in ₹).

- (a) 21120 (b) 20120 (c) 21150 (d) 18150

Q.273. The ratio of marks obtained by Rajesh, Rakesh and Ramesh in an exam is 2 : 4 : 9. What are the marks obtained



by Rakesh and Ramesh, if Rajesh scored 30 marks in the exam?

- (a) Rakesh = 40, Ramesh = 90
(b) Rakesh = 60, Ramesh = 135
(c) Rakesh = 120, Ramesh = 180
(d) Rakesh = 90, Ramesh = 40

Q.274. By mistake, instead of dividing ₹702 among Ram, Ramesh, and Naresh in the ratio $\frac{1}{3} : \frac{1}{4} : \frac{1}{6}$, it was divided in the ratio of 3 : 4 : 6. Who gained the most and by how much?

- (a) Ram ₹158 (b) Ramesh ₹158
(c) Naresh ₹168 (d) Ram ₹168

Q.275. Which smallest positive number should be subtracted from each of 9 and 13 so that 18 is the third proportion to them?

- (a) 2 (b) 4 (c) 3 (d) 1

Q.276. A varies directly as the positive square root of B, and inversely as the cube of C. If A = 15, when B = 27 and C = 2, then find B when A = 9 and C = 2.

- (a) $\frac{281}{42}$ (b) $\frac{243}{25}$ (c) $\frac{264}{37}$ (d) $\frac{275}{51}$

Q.277. The mean proportional of 6 and 54 is _____ more than 15.

- (a) 3 (b) 6 (c) 4 (d) 5

Q.278. Four numbers are in the ratio 1 : 2 : 3 : 5. Two times their sum is 44. Find the ratio of the numbers obtained by subtracting 3 from the cube of each number.

- (a) 2 : 31 : 204 : 999
(b) 3 : 13 : 208 : 992
(c) 11 : 67 : 219 : 1003
(d) 5 : 61 : 213 : 997

Q.279. The salaries of A, B and C are in ratio of: $\frac{3}{2}, \frac{6}{5}, \frac{4}{3}$. The salary of A and B together is Rs.40,500. By what percentage is the salary of A more than that of C?

- (a) 12% (b) 13% (c) 11.5% (d) 12.5%

Q.280. The ratio of a father's age to his Mother's age is 3 : 2. The product of the numbers representing their age is 486. The ratio of their ages after 5 years will be:

- (a) 32:24 (b) 32:21 (c) 32:22 (d) 32:23

Q.281. Salaries of B, C, D and E are in the ratio of 2 : 3 : 4 : 5 respectively. Their salaries are increased by 20 percent, 30 percent, 40 percent and 50 percent respectively. If the increased salary of D is ₹560, then what is the sum of the original salaries of B, C, D and E?

- (a) ₹1400 (b) ₹1560 (c) ₹1820 (d) ₹1260

Q.282. The cost of a diamond is directly proportional to the square of its weight. The cost of a 14 gm diamond is ₹2560. This diamond got broken down into two pieces in the ratio of 5 : 9. How much loss percent is incurred due to this breakage? (correct to two decimal places)

- (a) 53.47 percent (b) 49.71 percent
(c) 45.92 percent (d) 55.41 percent

Answer Key :-

1.(c)	2.(d)	3.(c)	4.(a)
5.(b)	6.(c)	7.(b)	8.(a)
9.(d)	10.(a)	11.(d)	12.(d)
13.(b)	14.(d)	15.(a)	16.(d)
17.(d)	18.(a)	19.(a)	20.(d)
21.(a)	22.(c)	23.(a)	24.(b)
25.(c)	26.(c)	27.(c)	28.(b)
29.(c)	30.(a)	31.(b)	32.(d)
33.(d)	34.(b)	35.(c)	36.(b)
37.(d)	38.(a)	39.(b)	40.(c)
41.(b)	42.(a)	43.(b)	44.(a)
45.(b)	46.(d)	47.(c)	48.(a)
49.(c)	50.(d)	51.(c)	52.(d)
53.(b)	54.(a)	55.(b)	56.(a)
57.(b)	58.(d)	59.(d)	60.(b)
61.(b)	62.(c)	63.(a)	64.(b)
65.(a)	66.(b)	67.(a)	68.(b)
69.(a)	70.(a)	71.(b)	72.(a)
73.(a)	74.(a)	75.(d)	76.(c)
77.(c)	78.(c)	79.(a)	80.(c)
81.(a)	82.(b)	83.(c)	84.(a)
85.(d)	86.(a)	87.(b)	88.(c)
89.(a)	90.(a)	91.(d)	92.(c)
93.(a)	94.(d)	95.(c)	96.(d)
97.(a)	98.(b)	99.(b)	100.(a)
101.(b)	102.(c)	103.(a)	104.(c)
105.(a)	106.(b)	107.(b)	108.(a)
109.(c)	110.(d)	111.(a)	112.(c)
113.(d)	114.(a)	115.(c)	116.(c)
117.(b)	118.(c)	119.(d)	120.(a)
121.(d)	122.(c)	123.(c)	124.(a)
125.(d)	126.(c)	127.(a)	128.(b)
129.(b)	130.(c)	131.(b)	132.(c)
133.(a)	134.(c)	135.(a)	136.(b)
137.(b)	138.(a)	139.(c)	140.(c)
141.(d)	142.(b)	143.(c)	144.(c)
145.(c)	146.(a)	147.(b)	148.(b)

149.(b)	150.(d)	151.(d)	152.(d)
153.(d)	154.(a)	155.(a)	156.(c)
157.(b)	158.(a)	159.(d)	160.(c)
161.(b)	162.(a)	163.(b)	164.(a)
165.(d)	166.(a)	167.(b)	168.(c)
169.(b)	170.(b)	171.(d)	172.(a)
173.(b)	174.(a)	175.(c)	176.(c)
177.(d)	178.(d)	179.(b)	180.(c)
181.(d)	182.(a)	183.(d)	184.(a)
185.(a)	186.(c)	187.(b)	188.(c)
189.(b)	190.(c)	191.(c)	192.(b)
193.(c)	194.(a)	195.(d)	196.(a)
197.(c)	198.(d)	199.(d)	200.(d)
201.(c)	202.(d)	203.(c)	204.(d)
205.(c)	206.(b)	207.(a)	208.(c)
209.(b)	210.(c)	211.(b)	212.(d)
213.(c)	214.(b)	215.(a)	216.(d)
217.(b)	218.(d)	219.(b)	220.(b)
221.(d)	222.(a)	223.(c)	224.(a)
225.(c)	226.(d)	227.(a)	228.(c)
229.(d)	230.(a)	231.(c)	232.(d)
233.(a)	234.(a)	235.(d)	236.(d)
237.(d)	238.(a)	239.(c)	240.(d)
241.(b)	242.(b)	243.(c)	244.(a)
245.(d)	246.(a)	247.(a)	248.(d)
249.(a)	250.(c)	251.(d)	252.(a)
253.(b)	254.(c)	255.(b)	256.(b)
257.(c)	258.(b)	259.(d)	260.(d)
261.(b)	262.(c)	263.(c)	264.(c)
265.(c)	266.(a)	267.(b)	268.(d)
269.(d)	270.(a)	271.(a)	272.(a)
273.(b)	274.(c)	275.(d)	276.(b)
277.(a)	278.(d)	279.(d)	280.(d)
281.(a)	282.(c)		

Solutions :-

Sol.1.(c) Let ₹5 coins = 2x, ₹2 coins = 3x, and ₹1 coins = 5x.

According to the question,

Total amount

$$= (5 \times 2x) + (2 \times 3x) + (1 \times 5x)$$

$$\Rightarrow 252 = 10x + 6x + 5x$$

$$\Rightarrow x = \frac{252}{21} = 12$$

Now, the number of coins of each type is

$$\text{₹5 coins} = 2x = 2 \times 12 = 24$$

$$\text{₹2 coins} = 3x = 3 \times 12 = 36$$

$$\text{₹1 coins} = 5x = 5 \times 12 = 60$$

Sol.2.(d)



From:- $9A = 10B \Rightarrow \frac{A}{B} = \frac{10}{9}$

From:- $12B = 13C \Rightarrow \frac{B}{C} = \frac{13}{12}$

Ratio:- $A : B : C$
 $10 : 9 : 12$

$A : B : C \rightarrow 10 \times 13 : 9 \times 13 : 9 \times 12$

Now,

$A : C = 10 \times 13 : 9 \times 12 = 65 : 54$

Sol.3.(c)

Required ratio: $8 \times \frac{120}{100} : 7 \times \frac{115}{100}$

$= 9.6 : 8.05$

Sol.4.(a)

Fund after charges: ₹1500 - ₹250 = ₹1250

20% bonus: $₹1250 \times \frac{120}{100} = ₹1500$

A's share = $\frac{7}{12} \times 1500 = ₹875$

Sol.5.(b) Let the number of horses = H and pigeons = P

According to question,

$H + P = 200$ ----(i)

$2P + 4H = 580$

$P + 2H = 290$ ----(ii)

Equation (ii) - Equation (i)

$H = 290 - 200 = 90$

Number of pigeons = $200 - 90 = 110$

Sol.6.(c) 16 cookies = 4 cups sugar

24 cookies = $\frac{4}{16} \times 24 = 6$ cups

Sol.7.(b) Let marbles initially in each bag = $2x, 3x, 4x$

After adding 12 marbles to each bag

Marbles in each bag = $8y, 11y, 14y$

According to the question,

$2x + 12 = 8y$

$8y - 2x = 12$ ----(i)

And, $3x + 12 = 11y$

$11y - 3x = 12$ ----(ii)

By the equation (i) $\times 3$ - (ii) $\times 2$ we get;

$24y - 6x - (22y - 6x) = 36 - 24$

$2y = 12 \Rightarrow y = 6$

Putting the value of y in equation (i) we get;

$8 \times (6) - 2x = 12$

$2x = 48 - 12 = 36 \Rightarrow x = 18$

Now, total marbles initially has

$= (2x + 3x + 4x) = 9x \Rightarrow 9 \times 18 = 162$

Sol.8.(a) Here two values of function is given hence it is of two variable.

According to the question,

$f \propto x$ and $f \propto \frac{1}{y}$

$\Rightarrow f = k_1 x + k_2 \times \frac{1}{y}$... (i)

Where k_1 and k_2 are constant

On putting $x = 0, y = 1, f = 15$ in (i) we get

$f = k_1 x + k_2 \times \frac{1}{y}$

$k_2 = 15$... (ii)

On putting $x = 1, y = 15, f = 2$ in (i) we

get, $f = k_1 x + k_2 \times \frac{1}{y}$

$2 = k_1 + 15 \times \frac{1}{15}$... [from (ii)]

$k_1 = 1$. Now,

when $x = 5$ and $y = 3$

$\Rightarrow f = 1 \times 5 + 15 \times \frac{1}{3}$

$\Rightarrow f = 5 + 5 = 10$

Sol.9.(d) Let ₹10, ₹5, ₹2 coins be x, y, z respectively.

According to question,

$x + y + z = 25$ (i)

$10x + 5y + 2z = 120$... (ii)

Multiply by 2 in equation (i) and subtracting equation (i) from (ii) we get

$8x + 3y = 70$

Now on satisfying the options we find that only option (d) gives positive value of y and no. of coins must be positive.

Hence, no. of ₹10 coins = 8

Sol.10.(a) Mean proportional of 1.21 and

$0.09 = \sqrt{1.21 \times 0.09} = 0.33$

Mean proportional of 0.16 and 0.36

$= \sqrt{0.16 \times 0.36} = 0.24$

Required ratio = $0.33 : 0.24 = 11 : 8$

Sol.11.(d) Mean proportion

$= \sqrt{8a^2 \times \frac{18}{a^4}} = \sqrt{\frac{144}{a^2}} = \frac{12}{a}$

Sol.12.(d) Let ₹5, ₹2, ₹1 coins be x, y, z respectively.

According to question,

$x + y + z = 40$ (i)

$5x + 2y + z = 100$... (ii)

On subtracting equation (i) from (ii)

$4x + y = 60$

Now on satisfying the options we find

that only option (d) gives positive value of y and no. of coins must be positive.

Hence, no. of ₹5 coins = 13

Sol.13.(b) For the fourth proportion

$\frac{\frac{1}{3} \times 27}{\frac{4}{5} \times 30} = \frac{\frac{3}{7} \times 35}{x}$

$\Rightarrow \frac{9}{24} = \frac{15}{x} \Rightarrow x = 40$

Sol.14.(d)

Ratio of their concentration = $10 : 20 : 30$

Ratio of their quantity = $2 : 3 : x$

$\therefore$ Concentration of mixed solution

$= \frac{10 \times 2 + 20 \times 3 + 30 \times x}{5 + x} = 23$

$20 + 60 + 30x = 115 + 23x \Rightarrow 7x = 35, x = 5$

Sol.15.(a) Total students = 720

No. of Boys = $\frac{3}{8} \times 720 = 270$

No. of Girls = $\frac{5}{8} \times 720 = 450$

According to question

$\frac{270 + x}{450 + 18} = \frac{2}{3}$

$= 810 + 3x = 2 \times 468 \Rightarrow 3x = 936 - 810$

$x = \frac{126}{3} = 42$

Sol.16.(d) According to the question,

Ram = $\frac{12}{5} = 2.4$, Shyam = $\frac{18}{9} = 2$

Preetham = $\frac{4}{8} = 0.5$, Vinay = $\frac{10}{7} = 1.42$

Sol.17.(d) Let the number of plastic, wood and glass = $2x, 3x$ and $5x$

According to question,

$= 5x - 3x = 144 \Rightarrow x = 72$

Number of plastic woods = $2x = 144$

Now,

After increasing 25% = $144 \times \frac{5}{4} = 180$

Sol.18.(a) Fourth proportional

$= \frac{3^{12} \times 4^{12}}{12^{35}} \Rightarrow \frac{3^{12} \times 4^{12}}{3^{35} \times 4^{35}} = 3^{-23} \times 4^{-23} = 12^{-23}$

Sol.19.(a) According to the question,

$\frac{x+5}{2x+5} = \frac{2}{3} \Rightarrow x = 5$

Smallest number (x) = 5

Sol.20.(d) Given, $y \propto (x + 3)$

$y = k_1(x + 3)$ (I)

When $y = 8$, and $x = 1$

$\Rightarrow 8 = k_1 \times 4 \Rightarrow k_1 = 2$

$x \propto \frac{1}{z}$

$x = k_2 \times \frac{1}{z}$ (II)

$x = 2$ and $z = 6$.

$\Rightarrow 2 = k_2 \times \frac{1}{6} \Rightarrow k_2 = 12$

at $k_2 = 12$ equation (II) become

$x = \frac{12}{z}$, put this value in eq. (I)

$y = 2\left(\frac{12}{z} + 3\right) \Rightarrow y = 2\left(\frac{12 + 3z}{z}\right)$

$\Rightarrow y \times z = 24 + 6z \Rightarrow z(y - 6) = 24$

Sol.21.(a) $A : B = \frac{1}{3} : \frac{1}{5} = 5 : 3$

$B : C = \frac{1}{7} : \frac{1}{8} = 8 : 7$

$C : D = \frac{1}{10} : \frac{1}{21} = 21 : 10$



$$\begin{array}{l} A : B : C : D \\ 5 : 3 : 3 : 3 \\ 8 : 8 : 7 : 7 \\ 21 : 21 : 21 : 10 \end{array}$$

$$40 : 24 : 21 : 10$$

Sol.22.(c) According to the question,

$$x + y = 96 \text{ (let } x < y \text{)}$$

$$\text{So, } \frac{x+3}{96-x} = \frac{11}{20} \Rightarrow 20x + 60$$

$$= 96 \times 11 - 11x$$

$$31x = 996 \Rightarrow x = \frac{996}{31} = 32.12$$

Largest number (y)

$$= 96 - x = 96 - 32.12 \approx 63.87$$

Sol.23.(a) Given :-

$$5 : 3\frac{1}{3}, 6\frac{1}{4} : 3\frac{1}{8} \text{ and } 3.5 : 4.5$$

$$5 : 3\frac{1}{3} = 5 : \frac{10}{3} \Rightarrow 3 : 2 = 1.5$$

$$6\frac{1}{4} : 3\frac{1}{8} = \frac{25}{4} : \frac{25}{8} \Rightarrow 2 : 1 = 2$$

$$3.5 : 4.5 = 35 : 45 \Rightarrow 7 : 9 = 0.78$$

$$\text{Required ascending order} = 0.78 < 1.5 < 2$$

$$\text{OR } 3.5 : 4.5 < 5 : 3\frac{1}{3} < 6\frac{1}{4} : 3\frac{1}{8}$$

Sol.24.(b)

Let income of P, Q, and R is $9x$, $15x$ and $10x$
and savings of P, Q, and R is $10y$, $15y$, $12y$
According to the question,

$$10x - 12y = 10x \times \frac{60}{100}$$

$$10x - 12y = 6x \Rightarrow \frac{x}{y} = \frac{3}{1}$$

Ratio of expenditure of P, Q, and R

$$= (9x - 10y) : (15x - 15y) : (10x - 12y)$$

$$= (27 - 10) : (45 - 15) : (30 - 12)$$

$$= 17 : 30 : 18$$

Sol.25.(c) Fourth proportional

$$= \frac{(a^3 + b^3) \times (a^2 - b^2)}{(a + b)^2 \times (a^3 - b^3)}$$

$$= \frac{(a + b)(a^2 - ab + b^2) \times (a + b)(a - b)}{(a + b)^2(a - b)(a^2 + ab + b^2)}$$

$$= \frac{(a^2 - ab + b^2)}{(a^2 + ab + b^2)}$$

Sol.26.(c) According to the question,

$$(A \times 6\% + B \times 4\%)$$

$$= (A \times 7\% + B \times 11\%) \times \frac{2}{5}$$

$$A \times 30\% + B \times 20\% = A \times 14\% + B \times 22\%$$

$$A \times 16\% = B \times 2\% \Rightarrow A : B = 1 : 8$$

Sol.27.(c)

$$\text{Ratio} \rightarrow A : B$$

$$\text{Initial} \rightarrow 8x : 5x = 13$$

$$\text{Final} \rightarrow 5x : 8x = 13$$

According to the question,

$$\text{Difference in A's share } 8x - 5x = 300$$

$$x = \frac{300}{3} = 100$$

$$\text{Total sum} = (8x + 5x) = 13x = ₹1300$$

Sol.28.(b)

$$\text{Required number} = \frac{ad - bc}{(a + d) - (b + c)}$$

$$= \frac{8 \times 9 - 6 \times 2}{(8 + 9) - (6 + 2)} = \frac{72 - 12}{9} = \frac{20}{3}$$

$$\text{Now, } (3P - 6) = 14 \text{ and } (9P - 4) = 56$$

$$\text{Mean proportional} = \sqrt{14 \times 56} = 28$$

Sol.29.(c) A : B : C = 3 : 2 : 5

Difference between share of C and B

$$= 3 \text{ units} \Rightarrow 3 \text{ units} = ₹ 300$$

$$\text{Share of A more than B} = 1 \text{ units} = ₹ 100$$

Sol.30.(a) Mean proportional

$$= \sqrt{(a + b)(a - b)^3 \times (a - b)(a + b)^3}$$

$$= \sqrt{(a - b)^4 \times (a + b)^4}$$

$$= (a - b)^2 \times (a + b)^2$$

Sol.31.(b)

Ratio - Maths : English : Chemistry

$$\text{Initial} - 6 : 4 : 7$$

$$\text{Final} - 7.2 : 5.6 : 11.2$$

$$\text{Required ratio} = 7.2 : 5.6 : 11.2 \\ = 9 : 7 : 14$$

Sol.32.(d)

$$\text{Ratio} - 0.5 ₹ : 0.25 ₹ : 0.20 ₹$$

$$\text{Coins} - 2 : 3 : 5$$

$$\text{Amount} - 1 : 0.75 : 1$$

$$\text{Required difference} = (1 - 1) = ₹ 0$$

Sol.33.(d)

$$\text{Blue}_{(\text{men})} : \text{Red}_{(\text{women})} = 3x : 7x ;$$

$$\text{Red}_{(\text{men})} : \text{Blue}_{(\text{women})} = 2y : y ;$$

$$\frac{3x + y}{7x + 2y} = \frac{35}{76}$$

$$\Rightarrow 228x + 76y = 245x + 70y \Rightarrow \frac{x}{y} = \frac{6}{17}$$

$$\text{Number of men} = 3x + 2y = 52$$

$$\text{Number of women} = 7x + y = 59$$

$$\text{Required ratio} = 52 : 59$$

Sol.34.(b)

$$\text{Ratio} \rightarrow P : Q$$

$$\text{Income} \rightarrow 40 : 16$$

$$\text{Expenditure} \rightarrow 18 : 10$$

$$\text{Savings} \rightarrow 11 : 3$$

Therefore, the ratio of the savings of P
And Q = 11 : 3

Sol.35.(c) % decrease (X) in the number
of units manufactured by company A

$$= \frac{12500 - 10625}{12500} \times 100$$

$$= \frac{1875}{12500} \times 100 = 15\%$$

% decrease (Y) in the number of units
manufactured by company

$$B = \frac{34,000 - 30,600}{34,000} \times 100$$

$$= \frac{3,400}{34,000} \times 100 = 10\%$$

$$\text{Required ratio} = 15 : 10 = 3 : 2$$

Sol.36.(b)

$$\text{Ratio} \rightarrow A : B : C$$

$$\text{No of officers} \rightarrow 5 : 4 : 1$$

$$\text{Average} \rightarrow (600) : (600+150) : (600+400)$$

So, the average of 40 officers $\rightarrow$

$$= 600 + \frac{(150 \times 4) + (400 \times 1)}{(5 + 4 + 1)}$$

$$= 600 + \frac{1000}{10} = 700$$

So, the monthly salary of an officer

$$= 700 \times 30 = ₹21,000$$

Sol.37.(d)

$$\text{Ratio} \rightarrow \text{Income} = \text{exp.} + \text{Savings}$$

$$A \Rightarrow 12 = 5 + 7 \Rightarrow 4 \times 5$$

$$B \Rightarrow 15 = 7 + 8 \Rightarrow 3 \times 4$$

Difference b/w their expenditure $\rightarrow$

$$(4 \times 5 \times 5) - (3 \times 4 \times 7) \text{ units} = 4500 \text{ Rs.}$$

$$16 \text{ units} = 4500 \text{ Rs.}$$

Income of B ($3 \times 4 \times 15$) units

$$= \frac{4500}{16} \times 180 = 50625 \text{ Rs.}$$

Sol.38.(a)

$$\text{Share of A} = \frac{18400}{46800} = \frac{46}{117} \text{ of the sum.}$$

Let total sum be $117a$, share of A = $46a$

$$\text{Share of A + D} = \frac{8}{13} \times 117a = 72a$$

$$\text{Share of D} = 72a - 46a = 26a$$

$$\text{Share of B + C} = \frac{5}{13} \times 117a = 45a$$

Given, share of B : share of C = 5 : 4

So, share of B = $25a$

and share of C = $20a$

x = difference of share of A and B

$$= 46a - 25a = 21a.$$

y = Difference of share of C and D

$$= 26a - 20a = 6a$$

$$x - y = 21a - 6a = 15a$$

$$117a = 46,800, a = 400$$

$$\text{So, } 15a = 15 \times 400 = 6000$$

Sol.39.(b) Let price of one pen = x

and price of one pencil = y

$$\text{So, } 5x + 3y = 44 \dots\dots(1)$$

$$3x + 5y = 36 \dots\dots(2)$$

On solving (1) and (2), we get

$$x = 7 \text{ and } y = 3$$

$$\text{So required ratio} = 7 : 3$$

Sol.40.(c)

$$\text{Denomination} \rightarrow ₹ 2 : ₹ 5 : ₹ 10$$

$$\text{Number of coins} \rightarrow 5 : 4 : 1$$

$$\text{Amount Ratio} \rightarrow 10 : 20 : 10$$

$$\text{Total Amount} = 40 \text{ unit} = 80 \text{ Rs}$$

$$1 \text{ unit} = 2 \text{ Rs.}$$

$$\text{Number of 5 rupees coin} = 4 \times 2 = 8$$

$$\text{Sol.41.(b) Number of boys} = \frac{5}{8}$$



$$\text{Boys Below 10 years} = \frac{5}{8} \times \frac{1}{3} = \frac{5}{24}$$

$$\text{Girls Below 10 years} = \frac{3}{8} \times \frac{2}{3} = \frac{6}{24}$$

$$\text{Students Below 10 years} = \frac{11}{24}$$

$$\text{Students Above 10 years} = \frac{13}{24}$$

$$\text{Total students} = 260 \times \frac{24}{13} = 480$$

$$\text{Number of boys} = 480 \times \frac{5}{8} = 300$$

Sol.42.(a) Go through option (a)

$$\frac{5+3}{7+5} = \frac{8}{12} = \frac{2}{3} \text{ (which satisfy)}$$

$$\frac{5-1}{7+3} = \frac{4}{10} = \frac{2}{5} \text{ (which satisfy)}$$

Sol.43.(b)

Ratio $\Rightarrow$ Income = expenditure + saving

$$A (80\%) \Rightarrow 5 = 4 + 1 \times 8$$

$$B (85\%) \Rightarrow 20 = 17 + 3 \times 3$$

$$C (75\%) \Rightarrow 4 = 3 + 1 \times 20$$

According to question, Difference of income of A and C (80 - 40) units $\Rightarrow$ (40 units) = 18000 Rs.

Then, Income of B (60 units) = 27000 Rs

Sol.44. (a)

$$\text{Ratio} \rightarrow A : B$$

$$\text{Last year Income} \rightarrow 4 : 3$$

Now,

Last Year : Present Year

$$A \rightarrow (3 : 4) \times 4 \times 5$$

$$B \rightarrow (5 : 6) \times 3 \times 3$$

Last Year : Present Year

$$A \rightarrow 60 : 80$$

$$B \rightarrow 45 : 54$$

According to the question,

$$(80 + 54) \text{ units} = 8.04 \text{ lakh}$$

$$1 \text{ unit} = 0.06 \text{ lakh}$$

$$\text{Last year income of B (45 units)}$$

$$= 45 \times 0.06 = 2.7 \text{ lakh}$$

Sol.45.(b)

$$\text{Fractional Value} \rightarrow 66\frac{2}{3}\% = \frac{2}{3}$$

$$\text{Ratio} \rightarrow A : B$$

$$\text{Income} \rightarrow 5 : 7$$

$$\text{Expenditure} \rightarrow 2 : 3$$

$$\text{Savings} \rightarrow 4000 : 5000$$

Now,

$$(5 \times 3) - (7 \times 2) \text{ units}$$

$$= (3 \times 4000) - (2 \times 5000)$$

$$\Rightarrow 1 \text{ unit} = 2000 \text{ Rs.}$$

$$\text{Total income of A and B} \rightarrow (5 + 7) \text{ units}$$

$$12 \text{ units} = 12 \times 2000 = 24000 \text{ Rs.}$$

Sol.46. (d)

$$\text{Ratio} \rightarrow \text{Divisor} : \text{Quotient} : \text{Remainder}$$

$$7 : 12$$

3 : 2

$$21 : 36 : 14$$

According to the question

$$14 \text{ unit} = 14 \Rightarrow 1 \text{ unit} = 1$$

$$21 \text{ unit (Divisor)} = 21$$

$$\text{Required remainder} = \frac{21 \times 36 + 14}{9} = 5 \text{ (Rem.)}$$

Sol.47.(c)

Let the age of shyam = 5 units and

age of Ravi = 3 units

According to the question

$$(5 + 3) \text{ unit} = 32 \text{ years} \Rightarrow 1 \text{ unit} = 4 \text{ years}$$

$$\text{Age of Shyam} = 5 \times 4 = 20 \text{ years and}$$

$$\text{Age of Ravi} = 3 \times 4 = 12 \text{ years}$$

$$\text{Now, } \frac{20+x}{12+x} = \frac{7}{5}$$

$$100 + 5x = 84 + 7x \Rightarrow x = 8 \text{ years}$$

Short Trick :-

$$\text{Ratio} \Rightarrow \text{Ravi} : \text{Shyam}$$

$$\text{Present} \Rightarrow 3 : 5$$

$$\quad \quad \quad +2 \quad \quad +2$$

$$\text{After} \Rightarrow 5 : 7$$

According to question,

Sum of present age (3 + 5) units = 32 yrs.

Then (2 units) = 8 yrs.

Sol.48.(a)

$$\text{notes worth} \rightarrow 5 : 2 : 1$$

$$\text{Amount} \rightarrow 15 : 2 : 3$$

$$\text{No. of notes} = \frac{\text{Total amount}}{\text{note's worth}}$$

Ratio of number of notes

$$= \frac{15}{5} : \frac{2}{2} : \frac{3}{1} = 3 : 1 : 3$$

Sol.49. (c) Actual ratio = 3 : 4 : 5 : 8

$$\Rightarrow (3 + 4 + 5 + 8) \text{ unit} = 4360$$

$$\Rightarrow 20 \text{ unit} = 4360 \Rightarrow 1 \text{ unit} = 218$$

$$A's \text{ share (3 unit)} = 3 \times 218 = 654$$

$$B's \text{ share (4 unit)} = 4 \times 218 = 872$$

$$C's \text{ share (5 unit)} = 5 \times 218 = 1090$$

$$D's \text{ share (8 unit)} = 8 \times 218 = 1744$$

$$\text{Mistaken ratio} = \frac{1}{3} : \frac{1}{4} : \frac{1}{5} : \frac{1}{8}$$

$$= 40 : 30 : 24 : 15$$

$$= (40 + 30 + 24 + 15) \text{ unit} = 4360$$

$$= 1 \text{ unit} = 40$$

$$A's \text{ share (40 unit)} = 40 \times 40 = 1600$$

$$B's \text{ share (30 unit)} = 30 \times 40 = 1200$$

$$C's \text{ share (24 unit)} = 24 \times 40 = 960$$

$$D's \text{ share (15 unit)} = 15 \times 40 = 600$$

$$\text{Clearly D get } 1744 - 600 = \text{Rs. } 1144 \text{ less}$$

Sol.50.(d)

$$\text{Ratio} \rightarrow A : B : C$$

$$7 : 12$$

$$8 : 5$$

$$14 : 24 : 15$$

According to the question

$$(15 - 14) = 1 \text{ unit} = 214$$

$$\text{Sum (x) } 53 \text{ unit} = (14 + 24 + 15) \times 214$$

= 11342

Sol.51.(c) Let the price of the articles be 400 and 500.

According to the question,

$$\frac{400 \times \frac{100+x}{100}}{500 \times \frac{100-30}{100}} = \frac{10}{7}$$

$$\Rightarrow \frac{4(100+x)}{5 \times 70} = \frac{10}{7}$$

$$\Rightarrow 400 + 4x = 500 \Rightarrow x = 25\%$$

Sol.52.(d) Initial Ratio :- 3x, 4x, 6x

A/Q

$$(3x + 25) : (4x + 15) : (6x + y) = 4 : 5 : 9$$

From red & blue;

$$\Rightarrow \frac{3x+25}{4x+15} = \frac{4}{5}$$

$$\Rightarrow 15x + 125 = 16x + 60 \Rightarrow x = 65$$

Now use x = 65 with blue & green;

$$\Rightarrow \frac{4x+15}{6x+y} = \frac{5}{9}$$

$$\Rightarrow \frac{260+15}{390+y} = \frac{5}{9} \Rightarrow 2475 = 1950 + 5y$$

$$\Rightarrow 5y = 525 \Rightarrow y = 105$$

Sol.53.(b) Let initial no of boys and girls is 5x and 3x respectively

$$A/Q, \Rightarrow \frac{5x+10}{3x} = \frac{2}{1}$$

$$\Rightarrow 5x + 10 = 6x \Rightarrow x = 10$$

$$\text{Number of students} = 10 \times (5 + 3) = 80$$

$$\text{Sol.54.(a)} \quad \frac{1}{2}A = \frac{4}{10}B = \frac{1}{8}C$$

$$\frac{1}{2 \times 4}A = \frac{4}{10 \times 4}B = \frac{1}{8 \times 4}C$$

$$\frac{A}{8} = \frac{B}{10} = \frac{C}{32} \Rightarrow A : B : C = 4 : 5 : 16$$

Sol.55.(b) According to question,

$$\text{Ratio} \rightarrow A : B : C$$

$$10 : 13 : 13$$

$$4 : 4 : 5$$

$$40 : 52 : 65$$

So, Ratio of A and C = 40 : 65 or 8 : 13

Sol.56.(a)

$$\frac{2}{5} : \frac{1}{4} : \frac{3}{5} : 1 \text{ or } 8 : 5 : 12 : 20$$

According to question,

$$\Rightarrow (8 + 5 + 12 + 20) = 45 \text{ units} = 5 \text{ m}$$

$$\Rightarrow 5 \text{ units} = \frac{5}{45} \times 5 = \frac{25}{45} = \frac{5}{9} \approx 0.556 \text{ m}$$

Sol.57.(b)

$$\text{Ratio} \rightarrow M : N : O : P$$

$$3 : 4 : 4 : 4$$

$$5 : 5 : 6 : 6$$

$$8 : 8 : 8 : 10$$

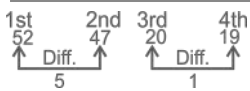
$$120 : 160 : 192 : 240$$

So,

the ratio of N and P = 160 : 240 or 2 : 3

Sol.58.(d)





Direct value of (x) =

$$\frac{5 \times (3\text{rd or } 4\text{th no.}) - 1 \times (1\text{st or } 2\text{nd no.})}{5 - 1}$$

$$\Rightarrow x = \frac{100 - 52}{4} = \frac{48}{4} = 12$$

Sol.59.(d) Let total red, blue and green marbles are respectively $2x$, $3x$, $5x$

After 10 red, 10 blue and m green marble added to the bag then marbles are respectively, $3y$, $4y$, $6y$

According to question,

$$2x + 10 = 3y \dots (i)$$

$$3x + 10 = 4y \dots (ii)$$

$$5x + m = 6y \dots (iii)$$

By eq. (i) $\times 3$ - and eq. (ii) $\times 2$ we get

$$6x + 30 = 9y$$

$$6x + 20 = 8y$$

$$- \quad - \quad -$$

$$y = 10 \text{ put eq. (i)}$$

$$x = 10$$

$$\text{Then, the value of } m = 6y - 5x$$

$$\Rightarrow m = 60 - 50 = 10$$

Sol.60.(b) P : Q : R : S

$$56 : 40 : 60 : 75$$

$$\text{Reqd. ratio} = 56 : 75$$

Sol.61.(b) Given: $x : y = 2 : 5$

$$(3x + 2y) : (7x - 2y)$$

$$(3 \times 2 + 2 \times 5) : (7 \times 2 - 2 \times 5)$$

$$16 : 4 = 4 : 1$$

Sol.62.(c) The marketing team's share of

$$\text{the total profit} = \frac{48000}{12} \times 4 = ₹16000$$

$$\text{After use remaining amount}$$

$$= 16000 - 4500 = ₹11500$$

The share of the top performer with a

$$\text{smaller portion} = \frac{11500}{5} \times 2 = ₹4600$$

Sol.63.(a) A : B : C : D

$$4 : 7 : 7 : 7$$

$$3 : 3 : 5 : 5$$

$$8 : 8 : 8 : 9$$

$$96 : 168 : 280 : 315$$

Sol.64.(b) Let number of ₹10, ₹5, and ₹2 coins are x , $2x$, $3x$ respectively

$$A/Q, 10 \times x + 5 \times 2x + 2 \times 3x = 416$$

$$\Rightarrow 10x + 10x + 6x = 416$$

$$\Rightarrow 26x = 416 \Rightarrow x = 16$$

$$\text{Number of coins are} \rightarrow 16, 32, 48$$

Sol.65.(a)

$$\frac{(3a + 4b)}{(3a - 4b)} = \frac{3 \times 5 + 4 \times 3}{3 \times 5 - 4 \times 3} = \frac{27}{3} \text{ or } 9 : 1$$

Sol.66.(b) According to the question,
20% of $a = 15\%$ of b

$$\frac{a}{b} = \frac{15\%}{20\%} = \frac{3}{4} \text{ or } 3 : 4$$

Sol.67.(a) Let the required number = x

Now, the ratio of the numbers

$5 - x, 8 - x, 7 - x, 12 - x$, respectively

$$\text{Therefore, } \frac{5 - x}{8 - x} = \frac{7 - x}{12 - x}$$

By going through the options, option (a) satisfied. Putting the value of $x = 2$.

Sol.68.(b)

$$\text{Ratio} \rightarrow W : X : Y : Z$$

$$3 : 5 : 5 : 5$$

$$5 : 5 : 7 : 7$$

$$7 : 7 : 7 : 9$$

$$105 : 175 : 245 : 315 \text{ or } 3 : 5 : 7 : 9$$

Sol.69.(a) A/Q,

$$\Rightarrow \frac{(P + Q)}{(P - Q)} = \frac{60}{40} = \frac{3}{2}$$

$$\Rightarrow \frac{2P}{2Q} = \frac{3+2}{3-2} = \frac{5}{1}$$

$$\Rightarrow P : Q = 5 : 1$$

Sol.70.(a) According to the Question:-

$$P : Q : R$$

$$150 : 120 : 100$$

$$\Rightarrow P : R = 3 : 2$$

Sol.71.(b) 20% of $(A + B) = 60\%$ of $(A - B)$

$$\Rightarrow (A + B) = (A - B) \times 3 \Rightarrow A + B = 3A - 3B$$

$$\Rightarrow 2A = 4B \Rightarrow A : B = 2 : 1$$

Sol.72.(a) ATQ, 44% Of $P = 24\%$ of Q

$$\frac{P}{Q} = \frac{24}{44} \Rightarrow P : Q = 6 : 11$$

Sol.73.(a) According to question,

$$\frac{20}{100} \times A = \frac{5}{10} \times B = \frac{1}{4} \times C$$

$$\frac{1}{5} \times A = \frac{1}{2} \times B = \frac{1}{4} \times C$$

$$A : B : C = 5 : 2 : 4$$

Sol.74.(a) $P \times 75 = Q \times 60 = R \times 30$

$$\Rightarrow P \times 5 = Q \times 4 = R \times 2$$

$$P : Q : R = 4 : 5 : 10$$

Sol.75.(d) According to question,

$$p \times \frac{3}{8} = q \times \frac{1}{4} \Rightarrow p : q = 2 : 3$$

Sol.76.(c) $A + B = 7$ ----- (i)

$A - B = 3$ ----- (ii)

On solving both equations

$$A = 5, B = 2$$

$$A^3 + B^3 = 5^3 + 2^3 = 125 + 8 = 133$$

$$A^3 - B^3 = 5^3 - 2^3 = 125 - 8 = 117$$

$$(A^3 + B^3) : (A^3 - B^3) = 133 : 117$$

Sol.77.(c) Ratio = $3 : 4 : 8$

According to question

$$\Rightarrow 8 \text{ units} = 40$$

$$\text{Total length} = 15 \times \frac{40}{8} = 75 \text{ cm}$$

Sol.78.(c) I : II : III

$$3 : 4 : 5 = 12$$

According to question,

$$\Rightarrow 12 \text{ Unit} = 15000$$

$$\Rightarrow (5 - 3) = 2 \text{ unit} = \frac{15000}{12} \times 2 = \text{Rs. } 2500$$

Sol.79.(a)

Ratio $\rightarrow$ First : Second

$$3 : 5$$

$$A/Q, (5 - 3) \text{ unit} = 16$$

$$1 \text{ unit} = 8$$

$$\text{Numbers are} \rightarrow 24, 40$$

Sol.80.(c) According to question,

$$\text{Weight} \rightarrow 4 : 5 : 6$$

$$+9 \quad +9 \quad +9$$

$$\text{New weight} \rightarrow 13 : 14 : 15$$

Hence, the given ratios themselves satisfy the condition,

$\therefore$ The original weight of the heaviest friend = 6 kg

Sol.81.(a) Given : $x : y = 3 : 7$

$$= \frac{(2x + 3y)}{(x + y)} = \frac{2(3) + 3(7)}{3 + 7}$$

$$= \frac{27}{10} \text{ or } 27 : 10$$

Sol.82.(b) According to question,

$$\begin{array}{ccc} \text{Acid} & : & \text{Water} \\ \text{Initial ratio} & \rightarrow & 2 : 1 \times 1 \\ & & 2 : 1 \end{array}$$

$$\begin{array}{ccc} \text{New ratio} & \rightarrow & 1 : 1 \times 2 \\ & & 2 : 2 \end{array} \quad \text{Diff. 1}$$

$$\Rightarrow (2 + 1) = 3 \text{ units} = 30 \text{ litres}$$

$$\Rightarrow 1 \text{ units} = 10 \text{ litres}$$

Sol.83.(c) $P : Q = 9 : 7$

$$9 \text{ units} = 27000 \Rightarrow 1 \text{ unit} = 3000$$

$$Q's \text{ share } (7) = 7 \times 3000 = \text{Rs. } 21000$$

Sol.84.(a)

$$P : Q : R : S$$

$$2 : 3 : 3 : 3$$

$$6 : 6 : 5 : 5$$

$$10 : 10 : 10 : 7$$

$$120 : 180 : 150 : 105$$

$$P : S = 120 : 105$$

$$= 8 : 7$$

Sol.85.(d) According to question,

$$10 \text{ units} \rightarrow 120$$

$$1 \text{ units} \rightarrow \frac{120}{10} = 12$$

After removing pencils:

$$\text{Red} = 3 \times 12 - 10 = 36 - 10 = 26$$

$$\text{Black} = 5 \times 12 - 20 = 60 - 20 = 40$$

$$\text{Required ratio} : 26 : 40 = 13 : 20$$

Sol.86.(a) 50% of $A = \frac{1}{4}$ of $B = \frac{1}{6}$ of C



$$\frac{1}{2}A = \frac{1}{4}B = \frac{1}{6}C$$

$$A : B : C = 2 : 4 : 6 = 1 : 2 : 3$$

Sol.87.(b)

$$\text{Weight} \rightarrow \begin{array}{ccc} 8 & 7 & 3 \\ +9 & +9 & +9 \end{array}$$

$$\text{New weight} \rightarrow 17 : 16 : 12$$

Hence, the given ratios themselves satisfy the condition,

$\therefore$ The original weight of the heaviest friend = 8 kg

Sol.88.(c)

$$\begin{array}{cccc} 1^{\text{st}} & 2^{\text{nd}} & 3^{\text{rd}} & 4^{\text{th}} \\ 52 & 47 & 20 & 19 \\ \uparrow & \uparrow & \uparrow & \uparrow \\ \text{Diff.} & & \text{Diff.} & \\ 5 & & 1 & \end{array}$$

Direct value of $(x) =$

$$\frac{5 \times (3^{\text{rd}} \text{ or } 4^{\text{th}} \text{ no.}) - 1 \times (1^{\text{st}} \text{ or } 2^{\text{nd}} \text{ no.})}{5 - 1}$$

$$\Rightarrow x = \frac{100 - 52}{4} = \frac{48}{4} = 12$$

$$\text{Now, } x + 5 = 12 + 5 = 17$$

Sol.89.(a) Let blue pens = m dozen

Black pens : Blue pens

$$\begin{array}{cc} 16 & m \\ 2 & 1 \end{array}$$

Total original price of pens = $32 + m$

If the number of dozens of the two colours had been interchanged

Then new price of the pens = $16 + 2m$

50% increases of his original price = $\frac{3}{2}$

$$\text{Now, } 16 + 2m = (32 + m) \cdot \frac{3}{2}$$

$$32 + 4m = 96 + 3m \Rightarrow m = 96 - 32 = 64$$

Hence, ratio of original pens, 16 : 64 or 1 : 4

Sol.90.(a) $P \times 75\% = Q \times 60\% = R \times 30\%$

$$= P \times 5 = Q \times 4 = R \times 2$$

$$P : Q : R = 4 : 5 : 10$$

Sol.91.(d) 4000 males $\rightarrow$ 3200 females

$$\text{for 1000 males} = \frac{3200}{4000} \times 1000 = 800$$

Sol.92.(c) Given : Ratio = 4 : 6 : 5

According to question,

$$\Rightarrow (4 + 6 + 5) = 15 \text{ units} = 150 \text{ marbles}$$

$$\Rightarrow 6 \text{ units} = \frac{150}{15} \times 6 = 60 \text{ marbles}$$

Sol.93.(a) Convert times to minutes:

$$\text{Jupiter : } 9 \text{ hr } 50 \text{ min} = 9 \times 60 + 50 = 590$$

$$\text{Saturn : } 11 \text{ hours } 40 \text{ minutes} = 11 \times 60 + 40 = 700$$

$$\text{Required ratio} = 590 : 700 = 59 : 70$$

Sol.94.(d) Saving of Rahim = 16000 - 12000 = ₹ 4000

$$\text{Saving of Ami} = \frac{204000}{12} - 12000 = ₹ 5000$$

$$\text{Required ratio} \rightarrow 4000 : 5000 \text{ or } 4 : 5$$

Sol.95.(c)

$$\text{Mean proportional} = \sqrt{9 \times 25} = 3 \times 5 = 15$$

$$\text{Sol.96.(d) Given: } \frac{21}{3} = \frac{7}{y}$$

$$\Rightarrow 21 \times y = 3 \times 7 \Rightarrow y = 1$$

Sol.97.(a) According to question,

$$\frac{10}{15} = \frac{15}{x} \Rightarrow x = \frac{15 \times 15}{10} = \frac{45}{2}$$

Sol.98.(b)

Let production of Maruti Alto = x and

production of Maruti Swift = y

Combined rejection rate:

$$\frac{5x + 9y}{x + y} = 8 \Rightarrow 5x + 9y = 8x + 8y$$

$$\Rightarrow \frac{y}{x} = \frac{3}{1} \text{ or } 3 : 1$$

Sol.99.(b) Let the total number of boys

in the school = $7x$ and the number of girls = $4x$

$$4x = 196 \Rightarrow x = 49$$

Total number of students in the college

$$= 11x = 11 \times 49 = 539$$

Sol.100.(a) Spain : Japan : France

$$\begin{array}{ccc} 3 & : & 4 \\ & & 6 : 5 \end{array}$$

$$\begin{array}{ccc} 18 & : & 24 : 20 \end{array}$$

Spain : France stamps = 18 : 20

$$= 9 : 10$$

Sol.101.(b) According to question

$$3A : 6B : 5C \Rightarrow A : B : C = 30 : 15 : 18$$

$$63 \text{ unit} = 630$$

$$C's \text{ share} = \frac{630}{63} \times 18 = 180 \text{ rupees}$$

Sol.102.(c) Flour : cocoa powder = 5 : 2

$$5 \text{ parts} = 500 \text{ gm}$$

$$1 \text{ parts} = \frac{500}{5} = 100 \text{ gm}$$

$$2 \text{ parts} = 100 \times 2 = 200 \text{ gm}$$

Sol.103.(a) Let the current number of teachers = x , and the number of students = 40x

According to the question,

$$\frac{40x + 60}{x + 4} = \frac{30}{1} \Rightarrow 40x + 60 = 30x + 120$$

$$10x = 60 \Rightarrow x = 6$$

Sol.104.(c) Given :- $36 : 8 = 9x : 16$

$$x = \frac{36 \times 16}{8 \times 9} = 8$$

Sol.105.(a) Let the two numbers be $3x$ and $7x$.

$$\Rightarrow 3x + 7x = 80 \Rightarrow x = 8$$

$$\text{Therefore, the numbers} = 3 \times 8 = 24,$$

$$7 \times 8 = 56$$

Sol.106.(b)

$$\text{Required ratio: } 8 \times \frac{115}{100} : 5 \times \frac{120}{100}$$

$$= 9.2 : 6 = 23 : 15$$

Sol.107.(b) According to question,

$$\Rightarrow \frac{32}{48} = \frac{64}{x} \Rightarrow x = 96$$

Sol.108.(a) Given :- Boys : Girls = 10 : 13

According to question,

$$\Rightarrow (10 + 13) = 23 \text{ units} = 115$$

$$\Rightarrow 10 \text{ units} = \frac{115}{23} \times 10 = 50$$

Sol.109.(c) Let , present age = $5x$, $6x$

$$\frac{5x + 15}{6x + 15} = \frac{8}{9} \Rightarrow 45x + 135 = 48x + 120$$

$$3x = 15 \Rightarrow x = 5$$

$$\text{Present age} = 25, 30$$

Sol.110.(d)

$$\text{Total income of P, Q, R} = 13 + 14 + 15 = 42$$

$$\text{Income of Q} = \frac{231000}{42} \times 14 = ₹ 77000$$

Sol.111.(a) Let age of three children are p year, q year, r year

According to question,

$$\text{Man's age (M)} = 4(p + q + r) \dots(i)$$

After 6 Years

$$\Rightarrow M + 6 = 2(p + 6 + q + 6 + r + 6)$$

$$\Rightarrow M + 6 = 2(p + q + r) + 36$$

From eq (i)

$$\Rightarrow 4(p + q + r) + 6 = 2(p + q + r) + 36$$

$$\Rightarrow 2(p + q + r) = 30 \Rightarrow (p + q + r) = 15$$

$$\therefore \text{Man's Present age} = 4 \times 15 = 60 \text{ years}$$

Sol.112.(c) Let number of votes got the contestants are $8x$, $7x$, $5x$

According to question,

$$\Rightarrow 7x = 4725 \Rightarrow x = 675$$

$$\therefore \text{Total number of valid votes (20x)}$$

$$= 20 \times 675 = 13,500$$

Sol.113.(d) Let the first part and second part be x and y respectively,

According to question,

$$\Rightarrow \frac{141 \times \frac{x}{8}}{141 \times \frac{y}{9}} = \frac{5}{6}$$

$$\Rightarrow \frac{9x}{8y} = \frac{5}{6} \Rightarrow \frac{x}{y} = \frac{20}{27}$$

$$\text{Now, } 47 \text{ units} = 141$$

$$\Rightarrow 20 \text{ units} = \frac{141}{47} \times 20 = 60$$

Sol.114.(a)

Ratio of A's and B's salaries = 2 : 3

According to question,

$$A's \text{ expenditure} = \frac{1}{2} \times A's \text{ salary}$$

$$\Rightarrow (2x - \text{saving of A}) = \frac{1}{2} \times 2x$$

$$\Rightarrow \text{Saving of A} = x = \text{Saving of B (given)}$$

$$\text{Expenditure of B} = 3x - x = 2x$$



$$\text{Required percentage} = \frac{2x}{3x} \times 100$$

$$= 66.67\%$$

Sol.115.(c) Let the distribution of toffee among Pushkal, Quadir and Rustam be $2x$, $3x$ and $4x$ respectively.

According to question,

$$\Rightarrow 9x = 360 \Rightarrow x = 40$$

Pushkal = 80, Quadir = 120 and Rustam = 160

$$\text{Then, Quadir} = 120 \times \frac{20}{100} = 24$$

$$\text{Remaining toffees of Quadir} = 120 - 24 = 96$$

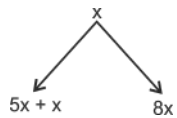
$$\text{Rustam} = 160 \times \frac{25}{100} = 40$$

$$\text{Remaining toffees of Rustam} = 160 - 40 = 120$$

$$\text{Pushkal has toffees} = 80 + 24 + 40 = 144$$

$$\text{Now, Pushkal : Quadir : Rustam} = 144 : 96 : 120 \text{ or } 6 : 4 : 5$$

Sol.116.(c) Let no of invalid votes are = x
According to question,



$$B = 6x, A = 8x$$

$$\Rightarrow (8x - 6x)\text{unit} = 28 \Rightarrow x = 14$$

$$\therefore \text{Total no of votes polled (14x)}$$

$$= 14 \times 14 = 196$$

Sol.117.(b) Mean proportional between p and $q = \sqrt{p \times q} = 8$

$$\text{Squaring on both sides} = p \times q = 64$$

By checking options, option (b) satisfies i.e. 16 and 4

Sol.118.(c)

Ratio - Rama : Ganga

Income - $4x$: $3x$

Expendi. - $3y$: $2y$

Savings - $4x - 3y$: $3x - 2y$

According to the question,

$$4x - 3y = 3x - 2y$$

$$x = y$$

savings of Rama ($4x - 3y$)

$$= 4x - 3x = x = ₹2700$$

So, expenditure of Rama = $3x = ₹8100$

And expenditure of Ganga = $2x = ₹5400$

Sol.119.(d)

Denomination $\rightarrow ₹0.5 : ₹0.2 : ₹0.1$

Number of coins $\rightarrow 2 : 3 : 4$

Amount Ratio $\rightarrow 1 : 0.6 : 0.4$

Total amount = $1 + 0.6 + 0.4 = 2$ unit

According to the question,

2 unit = ₹1020

So, number of 10 paise coins (4 unit)

$$= ₹1020 \times \frac{4}{2} = ₹2040$$

Sol.120.(a) Let x be the number added to each the four numbers 5, 19, 7 and 25

$$\text{Formula} \therefore x = \left| \frac{ad - bc}{(a + d) - (b + c)} \right|$$

$$\Rightarrow x = \frac{19 \times 7 - 5 \times 25}{(5 + 25) - (19 + 7)} = \frac{133 - 125}{30 - 26} = \frac{8}{4} \Rightarrow x = 2$$

Sol.121.(d) In 49 : $x :: x : 81$

$$x^2 = \sqrt{49 \times 81} \Rightarrow x = 63$$

In 64 : $y :: y : 169$

$$y^2 = \sqrt{64 \times 169} \Rightarrow y = 104$$

Now

$$2x + 3y = 2 \times 63 + 3 \times 104$$

$$= 126 + 312 = 438$$

Sol.122.(c)

$$\text{Formula} \therefore x = \left| \frac{ad - bc}{(a + d) - (b + c)} \right|$$

$$\Rightarrow x = \left| \frac{2 \times 35 - 3 \times 30}{(2 + 35) - (3 + 30)} \right|$$

$$= \left| \frac{-20}{4} \right| = |-5| = 5$$

Now, second largest part of the proportion = $(30 + 5) = 35$

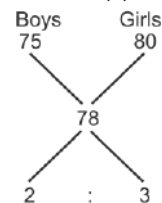
Sol.123.(c)

Required proportion = $30 : 24 :: 35 : x$

$$30 \times x = 24 \times 35$$

$$x = \frac{24 \times 35}{30} = 28 \text{ litres}$$

Sol.124.(a)



Required ratio = $2 : 3$

Sol.125.(d) According to the question,

$\{A = 3B, C = B + 5\}$ ----- (i)

$A + B + C = 75$ ----- (ii)

By putting the value of A and C from eq

(i) to (ii) we get;

$$3B + B + (B + 5) = 75$$

$$5B = 70 \Rightarrow B = 14 \Rightarrow A = 3B = 42$$

$$\text{Hence, } (B - 5) : (A + 3) = 9 : 45 = 1 : 5$$

Sol.126.(c)

According to the question, $x \propto \frac{1}{y}$

$$x = k \times \frac{1}{y} \quad \{\text{where } k \text{ is constant}\}$$

Given that $x = 2$ and $y = 6$ we get;

$$2 = k \times \frac{1}{6} \Rightarrow k = 12$$

Now, for $x = 3$

$$3 = 12 \times \frac{1}{y} \Rightarrow y = 4$$

Sol.127.(a) Let the age of Soham be x

Then, age of Ravi = $4x$

Age of Mayadevi = $65 + x$

$$\text{Age of Ram} = \frac{1}{3}(65 + x)$$

According to question,

$$\frac{1}{3}(65 + x) - 4x = 7$$

$$\frac{65}{3} + \frac{x}{3} - 4x = 7 \Rightarrow -\frac{11x}{3} = 7 - \frac{65}{3}$$

$$\frac{11x}{3} = \frac{44}{3} \Rightarrow x = 4$$

So, the age of Mayadevi = $65 + 4 = 69$ yrs

Sol.128.(b) According to question,

$$\frac{A}{C} = \frac{B}{D} \Rightarrow \frac{25}{30} = \frac{40}{D} \Rightarrow D = \frac{30 \times 40}{25} = 48$$

Sol.129.(b)

Denomination $\rightarrow ₹1 : ₹5 : ₹10$

No of coins $\rightarrow 3x : 5x : 7x$

Total amount in the bag

$$= 3x + 5 \times 5x + 10 \times 7x$$

$$= 3x + 25x + 70x = 98x$$

$$\text{Now, } 98x = 980 \Rightarrow x = 10$$

$$\text{No of coins of } ₹10 = 7 \times 10 = 70$$

Sol.130.(c) According to the question,

$(9 - 2)$ unit = ₹560

$$7 \text{ unit} = ₹560 \Rightarrow 1 \text{ unit} = ₹80$$

$$\text{Share of S} = 80 \times 3 = ₹240$$

Sol.131.(b) Let P's income = 1000

Q's income = 1200

R's income = 900

$$\text{Q's savings} = 1200 \times \frac{1}{4} = 300$$

$$\Rightarrow \text{Q's expenditure} = 1200 - 300 = 900$$

Ratio of expenditure

P : Q : R

12 : 15 : 8

$$\Rightarrow 720 : 900 : 480$$

Hence, P's savings = 280

Q's savings = 300

R's savings = 420

Ratio of savings = $280 : 300 : 420$

$$= 14 : 15 : 21$$

Sol.132.(c) $2x + x + 3x = 1200$

$$\Rightarrow 6x = 1200 \Rightarrow x = 200$$

Now,

$$3(3x) + 2(x) = 11x = 11 \times 200 = 2200$$

Sol.133.(a) Let male = $2x$

Female = $3x$

New ratio

$$\frac{2x + 200}{3x} = \frac{5}{6} \Rightarrow 4x + 400 = 5x$$

$$\Rightarrow x = 400 \Rightarrow \text{Female} = 3 \times 400 = 1200$$

Sol.134.(c) Mean proportional of 12.96

$$\text{and } 0.16 = \sqrt{12.96 \times 0.16}$$

$$= \sqrt{3.6 \times 3.6 \times 0.4 \times 0.4}$$

$$= 3.6 \times 0.4 = 1.44$$

Sol.135.(a) Let the number of gold coins and non gold coins be G and N respectively,

$$\Rightarrow G = \frac{1}{4} \times N \Rightarrow \frac{G}{N} = \frac{1}{4}$$



$$G = x \text{ and } N = 4x$$

According to the question,

$$\Rightarrow \frac{x+20}{4x} = \frac{2}{3} \Rightarrow 3x+60=8x \Rightarrow x=12$$

Gold coins = 12

Non gold coins = $4 \times 12 = 48$

Total number of coins in the collection

$$= 12 + 48 + 20 = 80$$

Sol.136.(b)

Ratio $\rightarrow$ Men : Women

$$\begin{array}{ccc} 5 & : & 3 \\ 5 & : & 4 \end{array} \quad \left. \begin{array}{c} 3 \\ 4 \end{array} \right\} 1$$

According to question

$$\Rightarrow 1 \text{ unit} = 150$$

No of men employee in company (5 unit)

$$= 5 \times 150 = 750$$

Sol.137.(b) According to the question,

$$y \times \frac{18}{100} + b \times \frac{24}{100} = 30$$

$$\Rightarrow 3y + 4b = 500 \dots\dots(i)$$

$$\text{And, } 3y = 2b + 200$$

$$\Rightarrow 3y - 2b = 200 \dots\dots(ii)$$

Subtract equation (i) from equation (ii), we get

$$6b = 300 \Rightarrow b = 50$$

Put the value of b in equation (i)

$$3y + 200 = 500 \Rightarrow 3y = 300 \Rightarrow y = 100$$

Required ratio

$$y^2 : b^2 = 100 \times 100 : 50 \times 50 = 4 : 1$$

Sol.138.(a) Mean proportion = $\sqrt{pq}$

$$\Rightarrow \frac{1}{5a} = \sqrt{4a^2 \times x}$$

Taking square both side,

$$\Rightarrow \frac{1}{25a^2} = 4a^2 \times x \Rightarrow x = \frac{1}{100a^4}$$

Sol.139.(c) Given, Price $\propto$ (Weight)²

Let the weights of four pieces be $4x, 3x, 2x, 1x$

$\therefore$ Total Price of four pieces

$$= 16x^2 + 9x^2 + 4x^2 + 1x^2 = 30x^2$$

Original weight of unbroken diamond

$$= 4x + 3x + 2x + 1x = 10x$$

Original price of unbroken diamond

$$= (10x)^2 = 100x^2$$

Difference between prices

$$= 100x^2 - 30x^2 = 70x^2 \Rightarrow 70x^2 = 63000$$

$$\Rightarrow 1x^2 = 900 \Rightarrow 100x^2 = ₹90,000$$

Sol.140.(c) According to the question,

$$\frac{5x-40}{6x+50} = \frac{8}{11} \Rightarrow 55x-440=48x+400$$

$$\Rightarrow 7x = 840 \Rightarrow x = 120$$

Hence, the number of girls

$$= 6 \times 120 = 720$$

Sol.141.(d) Mean proportional = $\sqrt{ab}$

$$\Rightarrow 30 = \sqrt{18 \times A}$$

Taking square both side,

$$\Rightarrow 900 = 18 \times A \Rightarrow A = 50$$

Sol.142.(b)

Ratio $\rightarrow$ P : Q

Expenditure $\rightarrow$ 36% of P : 56% of Q

$$\frac{\text{Expenditure of P}}{\text{Expenditure of Q}} = \frac{\frac{36}{100} \times \text{salary of p}}{\frac{56}{100} \times \text{salary of q}}$$

$$\frac{1.5}{1} = \frac{\frac{36}{100} \times \text{salary of p}}{\frac{56}{100} \times \text{salary of q}}$$

$$\frac{1.5}{1} = \frac{36 \times \text{salary of p}}{56 \times \text{salary of q}}$$

$$\frac{1.5 \times 56}{36} = \frac{\text{salary of p}}{\text{salary of q}}$$

$$\frac{\text{salary of p}}{\text{salary of q}} = \frac{7}{3} \text{ or } 7 : 3$$

$$\text{Sol.143.(c)} \quad \frac{\text{Circumference}}{\text{diameter}} = \frac{3}{5}$$

$$\frac{6}{\text{diameter}} = \frac{3}{5} \Rightarrow \text{Diameter} = 10 \text{ cm}$$

Radius = 5 cm

Sol.144.(c) Given :

$$a : b : c : d : e : f = 5 : 7$$

Now, $(3a + 5c + 11e) : (3b + 5d + 11f)$

$$(3 \times 5 + 5 \times 5 + 11 \times 5) : (3 \times 7 + 5 \times 7 + 11 \times 7)$$

$$(15 + 25 + 55) : (21 + 35 + 77)$$

$$95 : 133 \Rightarrow 5 : 7$$

Sol.145.(c)

Ratio $\rightarrow$ A : B : C

Income $\rightarrow$ 2 : 9 : 3

Expense $\rightarrow$ 3 : 9 : 5

Given condition, A's saving is half of his total income

To satisfy this condition,

multiply income ratio by 3

Income $\rightarrow$ 6 : 27 : 9

Expense $\rightarrow$ 3 : 9 : 5

Saving $\rightarrow$ 3 : 18 : 4

Sol.146.(a)

Ratio - 5 : 10 : 20

$$3 : 5 : 7$$

Final - 15 : 50 : 140

Total amount = $15 + 50 + 140 = 205$ units

205 units = ₹ 1845

(required worth) 15 units

$$= \frac{1845}{205} \times 15 = ₹ 135$$

Sol.147.(b)

Let the age of younger boy be x year

Then, age of older boy = $3x$

Now, according to the question,

$$(3x - 9) = 5(x - 9)$$

$$3x - 9 = 5x - 45 \Rightarrow 2x = 36 \Rightarrow x = 18$$

So, the present age of each is 18 yr and 54 yr.

Sol.148.(b)

Ratio - ₹2 : ₹5 : ₹10

Number of coins - $5x : 2x : 50 - 7x$

Total amount

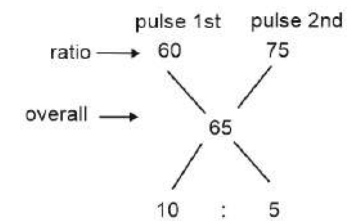
$$= 10x + 10x + 500 - 70x = 150$$

$$50x = 350, x = 7$$

So the number of 10 rupees coins

$$= 50 - (7 \times 7) = 1$$

Sol.149.(b)



Required ratio = 2 : 1

Sol.150.(d) According to the question,

$$M \propto \frac{1}{(N^2 + 3)}$$

$$M = k \times \frac{1}{(N^2 + 3)} \dots\dots(i)$$

{where k is constant}

Putting $M = 3$ and $N = 3$ in equation (i)

$$3 = k \times \frac{1}{(3^2 + 3)} \Rightarrow k = 36$$

Now, value of M when given that $N^2 = 69$

$$M = 36 \times \frac{1}{(69 + 3)} \Rightarrow M = \frac{36}{72} = \frac{1}{2}$$

Sol.151.(d) Let numbers be $7x$ and $3x$ respectively,

According to the question,

$$7x - 3x = 48 \Rightarrow x = 12$$

Sum of numbers = $10x = 120$

Sol.152.(d) 5 (Part of Ravi)

= 3 (Part of Reeta) = 11 (Part of Rahul)

$$\frac{\text{ravi}}{\text{reeta}} = \frac{3}{5}, \frac{\text{reeta}}{\text{rahul}} = \frac{11}{3}$$

Ratio - ravi : reeta : rahul

$$3 : 5 : 5$$

$$11 : 11 : 3$$

Final - 33 : 55 : 15

Sol.153.(d) According to the question,

$$\frac{3}{4} = 0.75, \frac{5}{8} = 0.625, \frac{1}{2} = 0.5,$$

$$\frac{6}{9} = 0.666$$

We can clearly see that the greatest ratio is 3 : 4

Sol.154.(a)

Ratio - a : b : c

$$3 : 7 : 7$$

$$4 : 4 : 9$$

Final - 12 : 28 : 63

$$\text{Sol.155.(a)} \quad \frac{8}{11}, \frac{4}{5}, \frac{1}{2} \text{ and } \frac{1}{11}$$

Denominators of given fractions are 11,



5, 2 and 11

LCM (11, 5, 2, 11) = 110

Multiply each fraction by 110 we get fractions

$$\frac{8}{11} \times 110, \frac{4}{5} \times 110, \frac{1}{2} \times 110, \frac{1}{11} \times 110$$

= 80, 88, 55 and 10

Here, 88 is the largest among all

Hence, required fraction = $\frac{4}{5}$

Sol.156.(c) Value of x =

$$\begin{aligned} & \left| \frac{ad-bc}{(a+d)-(b+c)} \right| \\ &= \left| \frac{7 \times 23 - 11 \times 18}{(7+23)-(11+18)} \right| \\ &= \left| \frac{161-198}{(30)-(29)} \right| = \left| \frac{-37}{1} \right| = 37 \end{aligned}$$

Then, mean proportion of $(x-1)$ and $(2x-10)$ = $\sqrt{(x-1)(2x-10)}$

$$\begin{aligned} &= \sqrt{(37-1)(74-10)} \\ &= \sqrt{36 \times 64} = 6 \times 8 = 48 \end{aligned}$$

Sol.157.(b) Required ratio = $\frac{28}{100} = 7 : 25$

Sol.158.(a) According to the question, Let the weight of the box and the gift articles is x and $9x$ gram respectively.

$$x + 9x = 25 \times 1000$$

$$10x = 25000 \Rightarrow x = 2500$$

So the weight of the articles will be

$$9 \times 2500 = 22500 \text{ gram}$$

Sol.159.(d) Third proportion

$$= \frac{(48)^2}{36} = 8 \times 8 = 64$$

Sol.160.(c)

Ratio $\rightarrow a : b : c : d$

$$3 : 4 : 4 : 4$$

$$2 : 2 : 3 : 3$$

$$5 : 5 : 5 : 6$$

$$30 : 40 : 60 : 72$$

Required ratio $\rightarrow 15 : 20 : 30 : 36$

Sol.161.(b) According to the question,

$$M : N : O = \left(\frac{1}{3} : \frac{1}{4} : \frac{1}{5} \right) \times 60$$

$$M : N : O = 20 : 15 : 12$$

Sol.162.(a) Mean proportion

$$= \sqrt{0.14 \times 2.24} = \sqrt{\frac{14}{100} \times \frac{224}{100}}$$

$$= \sqrt{\frac{14}{100} \times \frac{14 \times 16}{100}} = \frac{56}{100} = 0.56$$

Sol.163.(b) Let the fourth proportion be x

$$\Rightarrow \frac{1.25}{5.75} = \frac{3.5}{x} \Rightarrow x = \frac{3.5 \times 5.75}{1.25} = 16.1$$

Sol.164.(a) Sum of numbers = 5 unit

= 135

1 unit = 27

So, numbers = $2 \times 27 = 54$, $3 \times 27 = 81$

Sol.165.(d) Mean proportional = $\sqrt{ab}$

Mean proportional of 25 and 81

$$= \sqrt{25 \times 81} = 45$$

Sol.166.(a) Let third number = 100

Then, two number = $100 \times 130\% = 130$

and $100 \times 160\% = 160$

Ratio of two numbers = $130 : 160$

$$= 13 : 16$$

Sol.167.(b) Required ratio

$$= (640 - 400) : 400 = 240 : 400 = 3 : 5$$

Sol.168.(c) $15\% = \frac{3}{20}$, $25\% = \frac{1}{4}$

Sonu : Raju : Ramu

$$20 : 23$$

$$4 : 5 \Rightarrow 20 : 23 : 25$$

$$20 : 23 : 25$$

Required ratio = $23 : 25$

Sol.169.(b) Given :- $x : y = 5 : 6$

$$2x + 3y : 3x + 5y$$

$$= 2(5) + 3(6) : 3(5) + 5(6)$$

$$= 10 + 18 : 15 + 30 = 28 : 45$$

Sol.170.(b)

$$\frac{7}{4} = 1.7, \frac{8}{3} = 2.6, \frac{2}{3} = 0.6, \frac{1}{5} = 0.2$$

Clearly, we can see $\frac{8}{3}$ is the greatest of all the fractions given.

Sol.171.(d) No of red sweets in bag

$$= \frac{3}{4} \times 120 = 90$$

Sol.172.(a) $0.72 : x :: x : 2.85$

Product of extreme term = product of mean term

$$0.72 \times 2.85 = x \times x$$

$$x^2 = 2.052 \Rightarrow x = 1.4$$

Sol.173.(b) According to question,

$$\frac{5+x}{11+x} = \frac{3}{5}$$

$$25 + 5x = 33 + 3x \Rightarrow 2x = 8 \Rightarrow x = 4$$

Sol.174.(a) Let the no. be $3x$ and $7x$

According to question,

$$\frac{3x-11}{7x-11} = \frac{7}{20}$$

$$60x - 220 = 49x - 77 \Rightarrow 11x = 143 \Rightarrow x = 13$$

So, the smaller number = $3x = 3 \times 13 = 39$

Sol.175.(c)

$$40\% \text{ of } A = 60\% \text{ of } B = \frac{1}{6} \text{ of } C$$

$$\frac{2}{5} \text{ of } A = \frac{3}{5} \text{ of } B = \frac{1}{6} \text{ of } C = k \text{ (Let)}$$

$$A = \frac{5}{2}k, B = \frac{5}{3}k, C = 6k$$

$$A : B : C = \frac{5}{2}k : \frac{5}{3}k : 6k = 15 : 10 : 36$$

Sol.176.(c)

Ratio $\rightarrow A : B : C$

$$3 : 4$$

$$4 : 5$$

Final $\rightarrow 3 : 4 : 5$

$C - A = 5 - 3 = 2$ unit

2 unit = 30 marks

$$\text{Marks of B (4 unit)} = \frac{30}{2} \times 4 = 60 \text{ marks}$$

Sol.177.(d)

Ratio $\rightarrow$ Jeevan : Praveen : Kanan

$$5 : 8 :$$

$$3 : 2$$

Final $\rightarrow 15 : 24 : 16$

Share of Jeevan

$$= \frac{15}{15+24+16} \times 17,600$$

$$= \frac{15}{55} \times 17,600 = 4,800$$

Sol.178.(d)

Mean proportion = $\sqrt{75 \times 12} = 30$

Required natural number = $30 - 13 = 17$

Sol.179.(b) According to question,

$$a : b :: c : d \Rightarrow \frac{a}{b} = \frac{c}{d}$$

$$\text{Option (a) :- } b, a, d \text{ and } c, \Rightarrow \frac{b}{a} = \frac{d}{c}$$

$$\Rightarrow \frac{a}{b} = \frac{c}{d} \text{ (true)}$$

$$\text{Option (b) :- } c, d, b \text{ and } a, \Rightarrow \frac{c}{d} = \frac{b}{a}$$

$$\Rightarrow \frac{a}{b} \neq \frac{c}{d} \text{ (false)}$$

$$\text{Option (c) :- } d, c, b \text{ and } a, \Rightarrow \frac{d}{c} = \frac{b}{a}$$

$$\Rightarrow \frac{a}{b} = \frac{c}{d} \text{ (true)}$$

$$\text{Option (d) :- } a, c, b \text{ and } d, \Rightarrow \frac{a}{c} = \frac{b}{d}$$

$$\Rightarrow \frac{a}{b} = \frac{c}{d} \text{ (true)}$$

Sol.180.(c)

$$\text{Mean proportion} = \sqrt{ab} = \sqrt{a \times \frac{4}{a}} = 2$$

Sol.181.(d) $S = 75\%$ of T

$$\frac{S}{T} = \frac{75}{100} = \frac{3}{4} \Rightarrow S : T = 3 : 4$$

$$\text{Sol.182.(a)} \quad 20\% = \frac{1}{5}$$

Girls = 5 unit, Boys = 6 unit

required ratio of Boy to Girls = $6 : 5$

Sol.183.(d) 42% of Ranita's income = 56% of Bhaskar's income

$$\frac{\text{Income of Ranita}}{\text{Income of Bhaskar}} = \frac{56}{42} = \frac{4}{3}$$

According to question,

$$\frac{4x}{3x+200} = \frac{5}{4} \Rightarrow 16x = 15x + 1000$$

$$x = 1000$$

Hence, Ranita's Income
 $= 4 \times 1000 = 4000$

Sol.184.(a) $25\% = \frac{1}{4}$

Money left after saving

$$= \frac{3}{4} \times 40,000 = 30,000$$

Money spent on education

$$= \frac{2}{5} \times 30,000 = 12,000$$

$$\text{Required \%} = \frac{12000}{40000} \times 100 = 30\%$$

Sol.185.(a) No of students in high school

$$= \frac{324}{4} \times 3 = 243$$

Sol.186.(c) Ratio of boys and girls in class A = 11 : 18 = 0.61

Ratio of boys and girls in class

$$B = 13 : 17 = 0.76$$

Ratio of boys and girls in class

$$C = 20 : 9 = 2.22$$

$\Rightarrow$ class C has the biggest ratio of boys to girls.

Sol.187.(b)

Ratio $\rightarrow$ cheese : butter
 4 : 3

$$\Rightarrow 4 \text{ units} = 8$$

$$\Rightarrow 3 \text{ units} = \frac{8}{4} \times 3 = 6$$

So, she used 6 bowls of butter.

Sol.188.(c) S : F

Salaries $\rightarrow$ 2 : 3

Savings $\rightarrow$ 3 : 4

Since, total savings are equal to the salary of S.

$$\text{Total savings} = 3 + 4 = 7.$$

Now,

S : F

Salaries $\rightarrow$ 2 : 3

($\times 7$)

Savings $\rightarrow$ 3 : 4

($\times 2$)

Hence,

S : F

Salaries $\rightarrow$ 14 : 21

Savings $\rightarrow$ 6 : 8

Expenditure $\rightarrow$ 8 : 13

Sol.189.(b) $S \propto (R + 7)$

$$S = k(R + 7)$$

Putting, $S = 42$ and $R = 17$,

$$42 = k(17+7) \Rightarrow 42 = 24k \Rightarrow k = \frac{42}{24} = \frac{7}{4}$$

Now, putting $R = 29$ in the above expression, we have ;

$$S = k(R + 7) \Rightarrow S = \frac{7}{4}(29 + 7)$$

$$\Rightarrow S = \frac{7}{4} \times 36 = 63$$

Sol.190.(c) $8 - 7 = 1 \text{ unit} = ₹4800$

$$7 \text{ unit} = 4800 \times 7 = ₹33,600$$

So, the price of the washing machine
 $= ₹33,600$

Sol.191.(c) Let

Rupees with Pooja = $x + 50$ Rs.

Rupees with Chameli = x Rs.

Rupees with Anju = $2(x + 50)$ Rs.

Total Rupees = $x + 50 + x + 2x + 100$
 $= 450$ Rs.

$$\Rightarrow 4x + 150 = 450 \Rightarrow 4x = 300 \Rightarrow x = 75$$

Hence,

Pooja's Rupees = $75 + 50 = 125$ Rs.

Chameli's Rupees = 75 Rs.

Anju's Rupees = $2(75+50) = 250$ Rs.

Sol.192.(b) Let first number = $4x$

Second number = $5x$

According to question,

$$\frac{4x+3}{5x-7} = \frac{1}{1} \Rightarrow 4x+3 = 5x-7 \Rightarrow x = 10$$

Hence, First number = 40

And second number = 50

Sol.193.(c) Mean proportional

$$= \sqrt{45x^4 \times 5y^2} = 15x^2y$$

Sol.194.(a) According to the question,

$$\Rightarrow \frac{4x+8}{7x+8} = \frac{2}{3} \Rightarrow 12x+24 = 14x+16$$

$$\Rightarrow 2x = 8 \Rightarrow x = 4$$

Present age of A and B will be 16 yrs and 28 yrs respectively,

Now, sum of the age of A and B = $16 + 28 = 44$ years

Alternate method :-

	A	:	B	
Present Age $\rightarrow$	4	:	7	$\times 1$
	4	:	7	
Age after 8 yrs $\rightarrow$	2	:	3	$\times 3$ Diff. 2
	6	:	9	

$$\Rightarrow 2 \text{ units} = 8 \Rightarrow 1 \text{ unit} = 4$$

$$\Rightarrow 11 \text{ units} = 44 \text{ years}$$

Sol.195.(d) Given: $\frac{3+x}{11+x} = \frac{4+x}{14+x}$

By putting option one by one, we get
 $x = 1$

$$\text{As } \frac{3+1}{11+1} = \frac{4+1}{14+1} \Rightarrow \frac{5}{12} = \frac{5}{15}$$

Sol.196.(a) Let first number = $4x$

2nd number = $5x$

$$\text{Given } \frac{4x-17}{5x-17} = \frac{11}{14}$$

$$\Rightarrow 56x - 238 = 55x - 187 \Rightarrow x = 51$$

$$\Rightarrow \text{original ratio} = \frac{51 \times 4}{51 \times 5} = \frac{204}{255}$$

Hence,

$$\text{Required ratio} = \frac{204+20}{255+20}$$

$$= \frac{224}{275} \text{ or } 224 : 275$$

Sol.197.(c)

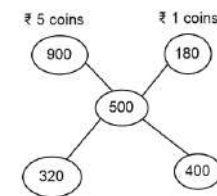
Let all coins are of denomination ₹5

$$\Rightarrow \text{Total value} = 180 \times 5 = 900 ₹$$

Let all coins are of denomination ₹1

$$\Rightarrow \text{Total value} = 180 \times 1 = 180$$

By allegation



$$\Rightarrow 4 : 5$$

$$\Rightarrow 9 \text{ unit} \rightarrow 180 \Rightarrow 1 \text{ unit} \rightarrow 20$$

Hence,

the number of ₹5 coins in the bag

$$= 4 \times 20 = 80.$$

Sol.198.(d) Let the third proportional = x

$$\frac{9}{7} = \frac{x}{14} \Rightarrow 9 \times 14 = 7x \Rightarrow x = 18$$

Sol.199.(d) Fourth proportional of 42, 56,

$$63 = \frac{63 \times 56}{42} = 84$$

Fourth proportional of 189, 273, 153

$$= \frac{153 \times 273}{189} = 221$$

Required ratio = 84 : 221

Sol.200.(d) According to the question,

$$\frac{9}{16}R = \frac{3}{4}S \Rightarrow \frac{R}{S} = \frac{3}{4} \times \frac{16}{9} = \frac{4}{3}$$

Now, $4 + 3 = 7 \text{ unit} = ₹4,760$

$$\text{Then, } 3 \text{ unit} = \frac{4760}{7} \times 3 = ₹2040$$

Sol.201.(c) Cyrus Rohan Mishti

3 : 2 : 2

7 : 7 : 12

21 : 14 : 24

According to the question,

$$24 - 21 = 3 \text{ unit} = ₹1800$$

$$\text{Then, } 14 \text{ unit} = \frac{1800}{3} \times 14 = ₹8400$$

Sol.202.(d) $(5^2 - 3^2) = 16, 5 - 3 = 2$

Third proportional to 16 and 2

$$= \frac{2 \times 2}{16} = \frac{1}{4}$$

Sol.203.(c) According to question,

$$b = 35 + a \text{ ----- (i)}$$

$$d = 60 + c \text{ ----- (ii)}$$

Since a,b,c,d are in proportion. Then we have ;

$$\frac{a}{b} = \frac{c}{d}$$

$$ad = bc$$

$$a(60 + c) = (35 + a)c \text{ [from eqn(i)&(ii)]}$$

$$60a + ac = 35c + ac \Rightarrow 60a = 35c$$

$$\frac{a}{c} = \frac{35}{60} = \frac{7}{12}$$

Let a and c be $7x$ and $12x$

$$\text{Then, } 7x \times 12x = 5376$$



$$84x^2 = 5376 \Rightarrow x^2 = \frac{5376}{84} = 64 \Rightarrow x = 8$$

Then, $a = 7 \times 8 = 56$, $b = 35 + 56 = 91$,
 $c = 12x = 12 \times 8 = 96$, $d = 60 + 96 = 156$
 So, $a + b + c + d = 56 + 91 + 96 + 156 = 399$

Sol.204.(d)

Types of coins ₹1 50P 25P
 No of coins 3 : 4 : 6

$$\text{Total amount} = 3 \times 1 + 4 \times \frac{1}{2} + 6 \times \frac{1}{4}$$

$$= 3 + 2 + 1.5 = 6.5$$

$$6.5 \text{ unit} = ₹143$$

$$4 \text{ unit} = \frac{143}{6.5} \times 4 = 88$$

Sol.205.(c) According to question,
 we have given ratio;

A	B	C	D
1	2	2	2
3	3	1	1
2	2	2	3

$$6 : 12 : 4 : 6 = 3 : 6 : 2 : 3$$

$$\text{Value of C} = \frac{2}{3+6+2+3} \times 875$$

$$= \frac{1750}{14} = 125$$

Sol.206.(b) Ratio of expenditure and
 saving = 2 : 1, So income = 3 unit

Now,

$$\text{Income} \rightarrow 3 \times 20\% = 60\%$$

$$\text{Expenditure} \rightarrow 2 \times 15\% = 30\%$$

$$\text{Saving} \rightarrow 1 \times ? = 30\%$$

$$\therefore ? = \frac{30}{1} = 30\%$$

% increase in her saving = 30%

Sol.207.(a)

Expenditure = price $\times$ consumption

Ratio of price = 3 : 4

Ratio of quantity = 4 : 7

$\therefore$ Ratio of expenditure = 12 : 28 or 3 : 7

Sol.208.(c)

$$A : B = 3 : 4$$

$$B : C = 12 : 13$$

$$A : B : C = 9 : 12 : 13$$

Here total amount = 34 unit = ₹ 7786

$\therefore$ B's amount = 12 unit

$$= \frac{7786}{34} \times 12 = ₹ 2748$$

Sol.209.(b)

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\text{Area} \rightarrow 4 : 3$$

$$\text{Height} \rightarrow 6 : 5$$

$$\text{Then ratio of bases} \rightarrow \frac{4}{6} : \frac{3}{5} = 10 : 9$$

Sol.210.(c) According to the question,

$$\frac{X}{Y} = \frac{3}{1} \text{ (put the value in equation)}$$

$$\frac{2X+3Y}{X-2Y} = \frac{2 \times 3 + 3 \times 1}{3 - 2 \times 1} = 9$$

$$\text{Sol.211.(b)} P \propto \frac{1}{Q} \Rightarrow P = k \times \frac{1}{Q}$$

$$10 = k \times \frac{1}{6} \Rightarrow k = 60$$

$$\text{now, } 15 = 60 \times \frac{1}{Q} \Rightarrow Q = 4$$

Sol.212.(d) Mean proportion

$$= \sqrt{(3 + 2\sqrt{2})^2 \times (3 - 2\sqrt{2})^2}$$

$$= (3 + 2\sqrt{2}) \times (3 - 2\sqrt{2}) \Rightarrow 9 - 8 = 1$$

Sol.213.(c) According to the question,

Ratio - A : B : C

Profit - $15x$: $6x$: $5x$

$$\text{B's share} = 29120 \times \frac{6x}{15x + 6x + 5x}$$

$$= ₹ 6720$$

Sol.214.(b)

Asha : Bhupinder : Gargi

$$9 : 5 : 5$$

$$2 : 2 : 1$$

$$18 : 10 : 5$$

Hence, shares of Asha

$$= \frac{66000}{(18 + 10 + 5)} \times 18 = ₹ 36,000$$

Sol.215.(a) According to the question,

$$5x - 2x = ₹ 150 \Rightarrow x = ₹ 50$$

$$\text{Hence, total sum} = (2x + 5x + 7x) = 14x$$

$$= ₹ 700$$

Sol.216.(d) $p : q = 5 : 2$,

$$\text{then } p^2 - q^2 : p^2 + q^2 = 25 - 4 : 25 + 4$$

$$= 21 : 29$$

Sol.217.(b)

Marks obtained by the student after

$$\text{studying for one hour} = \frac{60}{4} = 15 \text{ Marks}$$

So the time taken by him to study to get

$$75 \text{ marks} = \frac{75}{15} = 5 \text{ hours}$$

Sol.218.(d) According to the question

Let the numbers be $5x$, $2x$, $7x$

According to the question,

$$(5x)^2 + (2x)^2 + (7x)^2 = 3822$$

$$= 25x^2 + 4x^2 + 49x^2 = 3822$$

$$= 78x^2 = 3822 \Rightarrow x = 7$$

$$\text{So the largest no.} = 7x = 7 \times 7 = 49$$

Sol.219.(b) According to the question,

$$(19 + 6) - (7 + 15) \text{ units} = 72$$

$$3 \text{ units} = 72$$

Smallest number (6 units)

$$= \frac{72}{3} \times 6 = 144$$

Sol.220.(b) Let the savings of T = 1 unit

Then income of T = 8 units

Expenditure of T = $(8 - 1) = 7$ units

Hence, expenditure of T is 7 times of his savings

Sol.221.(d) Total students = 1650

Ratio $\rightarrow$ Boys : Girls

$$6 : 5$$

$$\text{Number of boys} = 1650 \times \frac{6}{11} = 900$$

$$\text{Number of girls} = 1650 \times \frac{5}{11} = 750$$

After Admission -

$$\text{Total boys} = 900 + 124 = 1024$$

New Ratio $\rightarrow$ Boys : Girls

$$4 : 3$$

$$4 \text{ unit} = 1024, 3 \text{ unit} = 768$$

$$\text{Newly admitted girls} = 768 - 750 = 18$$

$$\text{So, required ratio} = 18 : 124 = 9 : 62$$

Sol.222.(a) According to the question,

$$\frac{x+20}{2x} = \frac{1}{1} \Rightarrow x = 20$$

$$\text{Number of student initially} = 3x = 60$$

$$\text{Sol.223.(c)} \sqrt{A^2 + B^2} : \sqrt{C^2 - B^2}$$

$$= \sqrt{9 + 16} : \sqrt{25 - 16} \Rightarrow \sqrt{25} : \sqrt{9} = 5 : 3$$

Sol.224.(a)

$$3C + 5P = 1080 ₹ \dots \text{e.q. (1)}$$

$$2C + 4P = 840 ₹ \dots \text{e.q. (2)}$$

$$C + P = 240 ₹$$

$$\Rightarrow 2C + 2P = 480 ₹ \dots \text{e.q. (3)}$$

On solving e.q. (3) and e.q. (2), we get

$$2P = 360 \Rightarrow P = 180 ₹ \text{ and } C = 60 ₹$$

$$\text{Required ratio} = 1 : 3$$

Sol.225.(c) Compound ratio :-

$$\frac{3 \times 6}{4 \times 7} = \frac{36}{x} \Rightarrow x = 56$$

Sol.226.(d) Mean proportional (25) = $\sqrt{ab}$

$$\Rightarrow a = \frac{625}{b}$$

$$\text{Third proportional (25)} = \frac{b^2}{a}$$

$$\Rightarrow b^2 = 25a = 25 \times \frac{625}{b}$$

$$\Rightarrow b = 25 \text{ and } a = 25$$

$$\text{Sol.227.(a)} \frac{2x+8.5}{x+8.5} = \frac{7}{4}$$

$$x = 8.5 \times 3 = 25.5 \text{ years}$$

Sol.228.(c) Required ratio

$$= \left(\frac{23+7}{2}\right) : \left(\frac{23-7}{2}\right) = 15 : 8$$

$$\text{Sol.229.(d)} \frac{x+7}{2x+7} = \frac{3}{5}$$

$$x = 7 \times 2 = 14$$

$$\text{Greatest number (2x)} = 28$$

Sol.230.(a)

$$\text{Mean proportional} = \sqrt{7 \times 63} = 21$$

$$\text{Required difference} = 24 - 21 = 3$$



Sol.231.(c)

$$\frac{(6a^2 - 2b^2)}{(b^2 - a^2)} = \frac{(6 \times 25 - 2 \times 49)}{(49 - 25)}$$

$$= \frac{(150 - 98)}{(24)} = \frac{(52)}{(24)} = \frac{(13)}{(6)} = 13 : 6$$

Sol.232.(d)

$$\text{Fourth proportional} = \frac{3\sqrt{10} \times 5\sqrt{8}}{3\sqrt{5}}$$

$$= \sqrt{2} \times 5\sqrt{8} = 5 \times 4 = 20$$

Sol.233.(a)

Let their age be $5x$ and $3x$, then
 $5x \times 3x = 960 \Rightarrow x = 8$
 Their present age = 40 years and 24 years
 Required ratio = $40 + 6 : 24 + 6 = 23 : 15$

Sol.234.(a) Given numbers are in proportional, so $\frac{8}{P} = \frac{16}{18} \Rightarrow P = 9$

$$\text{Third proportional} = \frac{12 \times 12}{9} = 16$$

Sol.235.(d) Number of male worker = $960 - 420 = 540$
 Required ratio = $540 : 420 = 9 : 7$

Sol.236.(d) Given:- $x = 45$, $y = 9$, $x = Ky$
 $45 = K \times 9$, $K = 5$
 Now, $x = ?$, $y = 50$
 $x = 5 \times 50 = 250$

Sol.237.(d)

Ratio - X : Y : Z
 4 : 5 : 5
 7 : 7 : 6

Final - 28 : 35 : 30

$$\text{Share of Z} = \frac{1488}{(28 + 35 + 30)} \times 30 = ₹480$$

Sol.238.(a) Ratio :- Boys : Girls = 11 : 15
 difference between the number of boys and girls (4 units) = 200

$$\text{No. of boys (11 units)} = \frac{200}{4} \times 11$$

$$= 550 \text{ boys}$$

Sol.239.(c)

Ratio - Income = expenditure + savings
 Initial - 80 = 50 + 30
 Final - 96 = 65 + 31
 Change % in savings = $\frac{1}{30} \times 100$
 $= 3\frac{1}{3} \% \text{ increased}$

Sol.240.(d) 48% of male workers

$$= \frac{2}{3} \times \text{female worker}$$

$$\frac{\text{Male workers}}{\text{Female workers}} = \frac{2}{3} \times \frac{100}{48}$$

$$= \frac{100}{72} = \frac{25}{18}$$

So, required ratio :-

Male worker : female worker = 25 : 18

Sol.241.(b) Let the third number be 100.

According to question,
 The two numbers = 117 and 150
 Required ratio = 117 : 150 = 39 : 50

Sol.242.(b)

According to question, $\frac{96}{a} = \frac{c}{96}$

Going through all the options, while putting option (b)
 $a = 32$ and $c = 288$

$$\frac{96}{32} = \frac{288}{96} \Rightarrow 3 = 3 \text{ [satisfy]}$$

Sol.243.(c)

$$\text{Ratio of no.} = \frac{4}{5} : \frac{5}{6} : \frac{9}{10} = 24 : 25 : 27$$

Difference between greatest and smallest = $(27 - 24)$ units $\Rightarrow 3$ units = 12

$$\text{Middle number (25 units)} = \frac{12}{3} \times 25 = 100$$

Sol.244.(a) Let the length of colored saffron, white and green be $6x$, $7x$ and $8x$
 According to question,
 $6x + 7x + 8x = 9030 \Rightarrow x = 430$
 Length of green part = $8x = 8 \times 430$
 $= 3440 \text{ cm}$

Sol.245.(d) Let the number of storey be x
 According to question,
 $\frac{28}{7} = \frac{48}{x} \Rightarrow x = 12$

Sol.246.(a)

Let the number subtracted be x .

$$\frac{(2-x)}{(3-x)} = \frac{(3-x)}{4}$$

$$\Rightarrow 8 - 4x = 9 + x^2 - 6x$$

$$\Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x - 1)(x - 1) = 0$$

$$\Rightarrow (\text{smallest possible number}) x = 1$$

Sol.247.(a) Third proportional to $3x^2$ and

$$4xy = \frac{4xy \times 4xy}{3x^2} = \frac{16}{3} y^2$$

$$\frac{16}{3} y^2 = 48 \Rightarrow y^2 = \frac{48}{16} \times 3 = 9 \Rightarrow y = 3$$

Sol.248.(d)

Ratio $\rightarrow$ Cheese : sauce = 2 : 1

According to the question,
 Sauce used to make pizza (1 unit)
 $= 15$ cups

Cheese to be used (2 unit) = 30 cups

Sol.249.(a) Let total Income = 100 units

Then, Savings = 15 units
 Expenditure on health, education and food be 17 units, 34 units and 34 units respectively.

According to the question,
 $3 \times (34 \text{ units}) - 100 \text{ units} = 300 \text{ Rs.}$
 2 units = 300 Rs.

Required difference

$$= (17 \text{ units} - 15 \text{ units}) = 2 \text{ units} = 300 \text{ Rs.}$$

Sol.250.(c) $M = \sqrt{16 \times 9} = 12$

$$N = \frac{15^2}{9} = 25$$

$$5M + 3N = 5 \times 12 + 3 \times 25 = 60 + 75 = 135$$

Sol.251.(d) Ratio of manufacturing $\rightarrow$

Plate : Glasses = 2 : 1

and Glasses : Bowl = 2 : 1

Plate : Glasses : Bowl = 4 : 2 : 1

According to the question,

Total manufacturing (7 units) = 2380

Required difference (3 units)

$$= \frac{2380}{7} \times 3 = 1020 \text{ units}$$

Sol.252.(a) Let the numbers be $7x$ and $6x$

According to question,

$$(7x)^2 - (6x)^2 = 637$$

$$= 49x^2 - 36x^2 = 637$$

$$= x^2 = 49 \Rightarrow x = 7$$

Therefore, the numbers are 49 and 42.

Sol.253.(b) Ratio of Students $\rightarrow$

Initially = 2 : 3 : 5

Finally = 4 : 5 : 7

Increase in ratio (2 units) = 20 students

Initially total students (10 units) = 100 students

Sol.254.(c) Let the marks of practical and theory be 300 and 700.

According to question,

Required ratio

$$= (700 \times \frac{60}{100}) : (300 \times \frac{80}{100})$$

$$= 420 : 240 = 7 : 4$$

Sol.255.(b) $a : b : c = 4 : 7 : 9 \times 4$

$$\Rightarrow a : b : c = 16 : 28 : 36$$

$$\text{and } b : c : d = 28 : 36 : 21$$

$$\text{or } a : b : c : d = 16 : 28 : 36 : 21$$

Sol.256.(b) Let the number of boys and girls in the village be 500 and 300.

According to question,

Total number of boys and girls who appeared in the examination

$$\rightarrow (500 \times \frac{40}{100}) + (300 \times \frac{60}{100}) = 380$$

Total number of boys and girls who did not appear in the examination

$$\rightarrow 800 - 380 = 420$$

$$\text{Required ratio} = 380 : 420 = 19 : 21$$

Sol.257.(c) Volume of cylinder = $\pi r^2 h$

Ratio $\rightarrow$ Initial : Final

$$(\text{radius})^2 \rightarrow (2)^2 : (1)^2$$

$$\text{height} \rightarrow 5 : 2$$

$$\text{Volume} \rightarrow 20 : 2 = 10 : 1$$

Sol.258.(b) Let the numbers be $= 5x, 9x$ 

According to the question,

$$9x + 6 = 6(5x - 6)$$

$$9x + 6 = 30x - 36 \Rightarrow 42 = 21x \Rightarrow x = 2$$

So the required numbers = $5 \times 2 = 10$,

$$9 \times 2 = 18$$

Sol.259.(d) $\frac{J}{4} = \frac{K}{5} = \frac{L}{3}$

$$J = 4, K = 5, L = 3$$

According to the question,

$$(L + J)^2 : (J + K)^2 : (K + L)^2$$

$$(3 + 4)^2 : (4 + 5)^2 : (5 + 3)^2$$

$$49 : 81 : 64$$

Sol.260.(d) 1st no. : 2nd no. : 3rd no.

$$3 : 4 : 4$$

$$3 : 3 : 5$$

$$9 : 12 : 20$$

According to the question,

$$(9x \times 20x) \text{ unit} = ₹4500$$

$$x^2 = \frac{450}{18} \Rightarrow x = \sqrt{\frac{225}{9}} = 5$$

$$\text{Sum of three numbers} = (41x) = 41 \times 5 = 205$$

Sol.261.(b) Total amount = 50×80

$$\text{Number of children} = \frac{50 \times 80}{50} = 80$$

Sol.262.(c)

$$(x + y) : (y + z) : (z + x) = 3 : 5 : 7$$

$$\text{Let, } x + y = 3a, y + z = 5a, z + x = 7a$$

$$2(x + y + z) = 15a \text{ -----e.q.(1)}$$

$$45 = \frac{15a}{2} \Rightarrow a = 6$$

$$(x + y) = 3a = 18$$

$$\text{Now, } (x + y + z) = 45$$

$$18 + z = 45 \Rightarrow z = 27$$

Sol.263.(c) Let share of r be x.

Then, share of q and p = $(x+6)$ and $(x+15)$

According to question,

$$x + (x + 6) + (x + 15) = 120$$

$$3x = 99 \Rightarrow x = 33$$

$$\text{Required ratio} = 48 : 39 : 33 = 16 : 13 : 11$$

Sol.264.(c) According to the question,

Ratio $\rightarrow$ Income : expenditure : savings

$$\text{Initial} \rightarrow 7 : 4 : 3$$

$$\text{Change} \rightarrow -1.4 : +1.4$$

$$\text{Final} \rightarrow 7 : 2.6 : 4.4$$

His savings increased $\frac{1}{5}$ th of their total

income and income remains same

Therefore, the new ratio of his expenditure and savings = $2.6 : 4.4 = 13 : 22$

Sol.265.(c) Let, weight of Ayush and

Abhishek be $8x$ and $5x$ respectively.

Increased weight of Abhishek

$$= 5x \times \frac{140}{100} = 7x$$

Overall increased weight

$$\rightarrow (8x + 5x) \times \frac{160}{100} = 20.8x$$

According to the question,

$$20.8x = 104 \text{ kg.} \Rightarrow x = 5 \text{ kg.}$$

Increased weight of Ayush

$$\rightarrow 20.8x - 7x = 13.8x$$

$$13.8x = 13.8 \times 5 = 69 \text{ kg.}$$

Sol.266.(a) Let the income of Jitendra

and Narendra = $4x$ and $7x$

And expenditure of Jitendra and Narendra

= $6y$ and $5y$

According to the question,

$$4x = 2800 \rightarrow x = 700$$

$$\text{Savings of Jitendra} \Rightarrow 4x - 6y = 400$$

$$\Rightarrow y = 400$$

$$\text{Savings of Narendra} \Rightarrow 7x - 5y$$

$$= 4900 - 2000 = 2900 \text{ Rs.}$$

Sol.267.(b) Let income of P = x

According to the question;

$$\frac{1}{3}x + 4400 = \frac{1}{2}x \Rightarrow \frac{1}{6}x = 4400$$

$$x = \text{Rs. } 26,400$$

$$\frac{P}{Q} = \frac{1}{2} \Rightarrow Q = 26400 \times 2 = \text{Rs. } 52800$$

Sol.268.(d) Let the larger number be x

and the smaller number be y .

According to question,

$$x - y = \frac{20}{100} \times (x + y) \Rightarrow \frac{x - y}{x + y} = \frac{1}{5}$$

$$x - y = 1 \text{e.q.(1)}$$

$$x + y = 5 \text{e.q.(2)}$$

On solving e.q.(1) and (2); we get

$$x = 3 \text{ and } y = 2$$

$$\text{required ratio, } x : y = 3 : 2$$

Sol.269.(d) Total expenditure = 12000₹

According to the question,

$$\text{Expenditure of D} = \text{Saving of C} = x$$

$$\text{Then, Expenditure of C} = (12000 - x)$$

$$\text{Ratio} \rightarrow \text{C} : \text{D}$$

$$\text{Income} \rightarrow 3 : 4$$

$$\text{Expenditure} \rightarrow (12000 - x) : x$$

$$\text{Saving} \rightarrow x : (y)$$

$$\text{In 'C' case} \Rightarrow \text{Income} - \text{Expenditure}$$

$$= \text{Savings}$$

$$3 \text{ units} - (12000 - x) = x$$

$$\Rightarrow 1 \text{ unit} = 4000 \text{ ₹}$$

$$\text{Total Income (7 units)} = 28000 \text{ ₹}$$

Now,

$$\text{total saving} = 28000 - 12000 = 16000 \text{ ₹}$$

Short Trick :-

$$\text{Ratio} \rightarrow \text{C} : \text{D}$$

$$\text{Income} \rightarrow 3 : 4$$

$$\text{Expenditure} \rightarrow 2 : 1$$

$$\text{Saving} \rightarrow 1 : 3$$

$$\text{Total expenditure (3 units)} = 12000 \text{ ₹}$$

$$\text{Total Savings (4 units)} = 16000 \text{ ₹}$$

Sol.270.(a) Let current age of father be x and son be y

$$x + y = 94 \text{ -----e.q.(1)}$$

$$(8 - x) = 2(8 - y)$$

$$2y - x = 8 \text{ -----e.q.(2)}$$

On adding e.q.(1) and e.q.(2); we get

$$y = 34 \text{ and } x = 60$$

$$\text{Required ratio} = 70 : 44 = 35 : 22$$

Sol.271.(a) Ratio $\rightarrow$

$$P : Q + R = 4 : 7 \times 7$$

$$Q : P + R = 2 : 5 \times 11$$

Total amount of P, Q, R = 77 units

$$\text{And R's unit} = 77 - (7 \times 4 + 11 \times 2)$$

$$= 27 \text{ units}$$

$$77 \text{ units} = ₹43120$$

$$(R) 27 \text{ units} = \frac{43120}{77} \times 27 \rightarrow ₹15120$$

Sol.272.(a)

$$\text{Ratio} \rightarrow A : B$$

$$\text{Income} \rightarrow 7 : 4 \times 2$$

$$\text{Expenditure} \rightarrow 3 : 1 \times 3$$

$$\text{Savings of A} = \{(7 \times 2) - (3 \times 3)\}$$

$$= 5 \text{ units} \Rightarrow 5 \text{ units} = 4800 \text{ ₹}$$

Total Income of A and B

$$= (7 + 4) \times 2 = 22 \text{ units}$$

$$22 \text{ units} \rightarrow \frac{4800}{5} \times 22 = 21120 \text{ ₹}$$

Sol.273.(b) Ratio of their marks $\rightarrow 2 : 4 : 9$

According to the question,

Marks obtained by Rajesh (2 units) = 30

Marks obtained by Rakesh (4 units) = 60

Marks obtained by Ramesh (9 units) = 135

Sol.274.(c)

Ram : Ramesh : Naresh

$$\frac{1}{3} : \frac{1}{4} : \frac{1}{6} \Rightarrow 4 : 3 : 2$$

Ratio $\rightarrow$ Ram : Ramesh : Naresh

$$\text{Initial} \rightarrow 4 : 3 : 2 = 9 \text{ units} \times 13$$

$$\text{Final} \rightarrow 3 : 4 : 6 = 13 \text{ units} \times 9$$

Make units equal as money divided is same

Ram : Ramesh : Naresh

$$52 : 39 : 26 = 117 \text{ units}$$

$$27 : 36 : 54 = 117 \text{ units}$$

$$-25u \quad -3u \quad +28u$$

According to question,

$$117 \text{ units} = \text{Rs. } 702$$

$$\text{Then, } 28 \text{ units} = \text{Rs. } 168$$

Clearly, Naresh got Rs.168 More

Sol.275.(d) If c is the third proportion of

the number a and b , then $c = \frac{b^2}{a}$

Let, x be the no. which is subtracted from 9 and 13 so that 18 is the third proportional to them.

According to question,

$$\frac{(13 - x)(13 - x)}{(9 - x)} = 18$$

$$169 - 26x + x^2 = 162 - 18x$$

$$7 - 8x + x^2 = 0$$



$$x^2 - 7x - x + 7 = 0$$

$\Rightarrow (x - 7)(x - 1) = 0 \Rightarrow x = 7$ and $x = 1$
 Since, smallest positive number to be subtracted. Hence, $x = 1$

Sol.276.(b) $A \propto \sqrt{B} \propto \frac{1}{C^3}$

$$A = \frac{k \times \sqrt{B}}{C^3} \Rightarrow 15 = \frac{k \times \sqrt{27}}{2^3}$$

$$k = \frac{120}{3\sqrt{3}} \dots\dots\dots \text{e.q. (1)}$$

Again ,

$$A = \frac{k \times \sqrt{B}}{C^3} \Rightarrow 9 = \frac{120 \times \sqrt{B}}{3\sqrt{3} \times 2^3}$$

$$\sqrt{B} = \frac{9\sqrt{3} \times 8}{40} \Rightarrow B = \frac{243}{25}$$

Sol.277.(a) Mean proportion of 6 and 54

$$\Rightarrow \sqrt{6 \times 54} = 18$$

so, 18 is 3 more than 15.

Sol.278.(d)

Let the four no's be $x, 2x, 3x$ and $5x$

According to the question,

$$2(x + 2x + 3x + 5x) = 44$$

$$\Rightarrow 11x = 22 \Rightarrow x = \frac{22}{11} = 2$$

So, the no's are 2, 4, 6 and 10

$$\text{Required ratio} = 2^3 - 3 : 4^3 - 3 : 6^3 - 3 : 10^3 - 3 = 5 : 61 : 213 : 997$$

Sol.279.(d) LCM of (2, 5, 3) = 30

Then, the ratio of the salary of A, B and C

$$= \frac{3}{2} \times 30 : \frac{6}{5} \times 30 : \frac{4}{3} \times 30$$

$$= 45 : 36 : 40$$

$$\text{Required percentage} = \frac{45 - 40}{40} \times 100$$

$$= \frac{5}{40} \times 100 = 12.5\%$$

Sol.280.(d) Let the age of father and mother be $3x$ and $2x$ respectively

According to the question,

$$(3x) \times (2x) = 486$$

$$\Rightarrow 6x^2 = 486 \Rightarrow x^2 = \frac{486}{6} = 81$$

$$\Rightarrow x = \sqrt{81} = 9 \text{ yrs.}$$

Then, the age of father and mother

$$= 9 \times 3 = 27 \text{ yrs and } 2 \times 9 = 18 \text{ yrs.}$$

So, the ratio of the ages after 5 yrs.

$$= (27 + 5) : (18 + 5) = 32 : 23$$

Sol.281.(a)

Ratio $\rightarrow B : C : D : E$

Initial Salaries $\rightarrow 20 : 30 : 40 : 50$

Final salary $\rightarrow 24 : 39 : 56 : 75$

According to the question,

Increased salary of D = 56 unit

56 unit = ₹560

Sum of original salaries of B, C, D and E

$$= \frac{560}{56} \times 140 = ₹1,400$$

Sol.282.(c) It is given that, $CP \propto W^2$

Initial weight of a diamond = $(5 + 9)^2$

$$= 14^2 = 196 \text{ unit}$$

After break down, weight of a diamond

$$= 5^2 + 9^2 = 25 + 81 = 106 \text{ unit}$$

Required loss% in weight of diamond

$$= \frac{196 - 106}{196} \times 100$$

$$= \frac{90}{196} \times 100 = 45.918 \sim 45.92\%$$

